HYDRAULIC LASH ADJUSTER (HLA) REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5]

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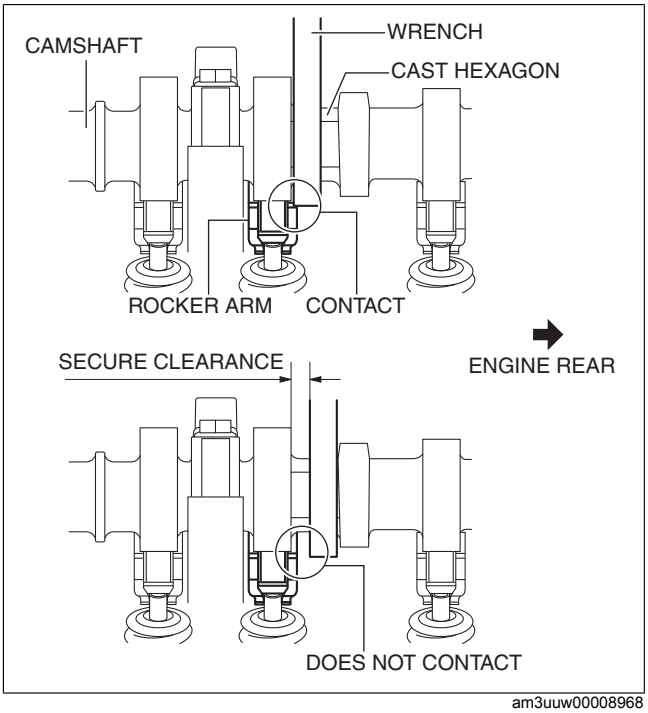
SKYACTIV-G 1.3, SKYACTIV-G 1.5 (with 4-1 exhaust system)

Warning

- **A hot engine can cause severe burns. Turn off the engine and wait until it is cool before servicing.**

Caution

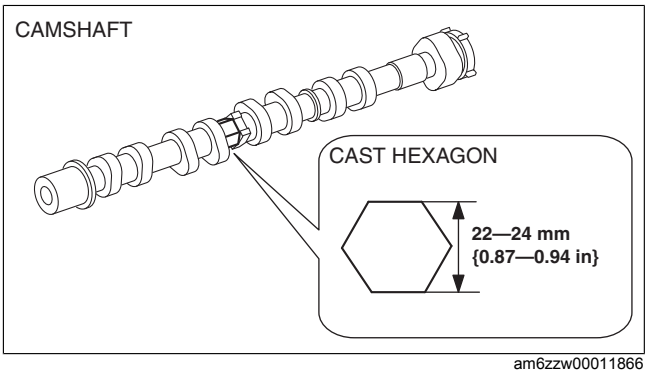
- **If the camshaft is rotated with the timing chain removed and the piston at the top dead center position, the valve may contact the piston and the engine could be damaged. When rotating the camshaft with the timing chain removed, rotate it after lowering the piston from the top dead center position.**
- **When rotating the camshaft using a wrench on the cast hexagon, the wrench may contact the rocker arm and damage the rocker arm. To prevent damage to the rocker arm when holding the camshaft on the cast hexagon, use a wrench on the rear side of the engine as shown in the figure to secure a clearance between the cam.**



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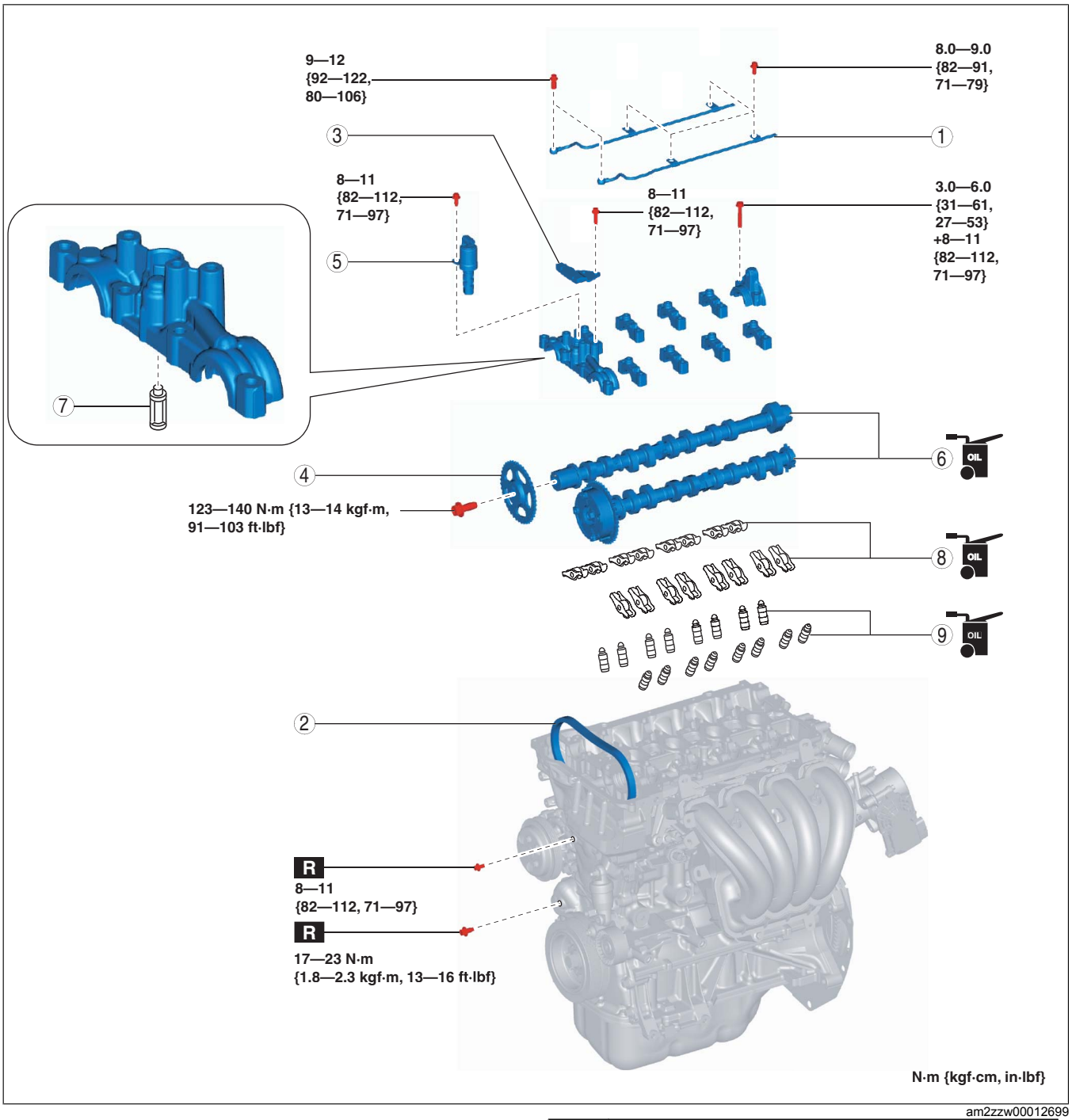
Note

- Width at the cast hexagon of the camshaft is **22—24 mm {0.87—0.94 in}**.



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1. Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.)
2. Remove the plug hole plate. (See PLUG HOLE PLATE REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
3. Remove the ignition coil.(See IGNITION COIL/ION SENSOR REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
4. Remove the cylinder head cover. (See TIMING CHAIN REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
5. Remove the vacuum pump. (See VACUUM PUMP REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
6. Remove the high pressure fuel pump and rear housing. (See HIGH PRESSURE FUEL PUMP REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
7. Remove in the order indicated in the table.
8. Install in the reverse order of removal.
9. Start the engine and inspect the following:
   - Ignition timing, idle speed and idle mixture. (See ENGINE TUNE-UP [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)



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1	Oil shower pipe (See Oil shower pipe installation note.)
2	Timing chain (See Timing chain removal note.) (See Timing chain installation note.)
3	Chain guide (No.1)
4	Exhaust camshaft sprocket (See Exhaust camshaft sprocket removal note.) (See Exhaust camshaft sprocket installation note.)
5	OCV
6	Camshaft (See Camshaft removal note.) (See Camshaft installation note.)

7	OCV oil filter
8	Rocker arm (See Rocker arm removal note.) (See Rocker arm installation note.)
9	HLA (See HLA removal note.) (See HLA installation note.)

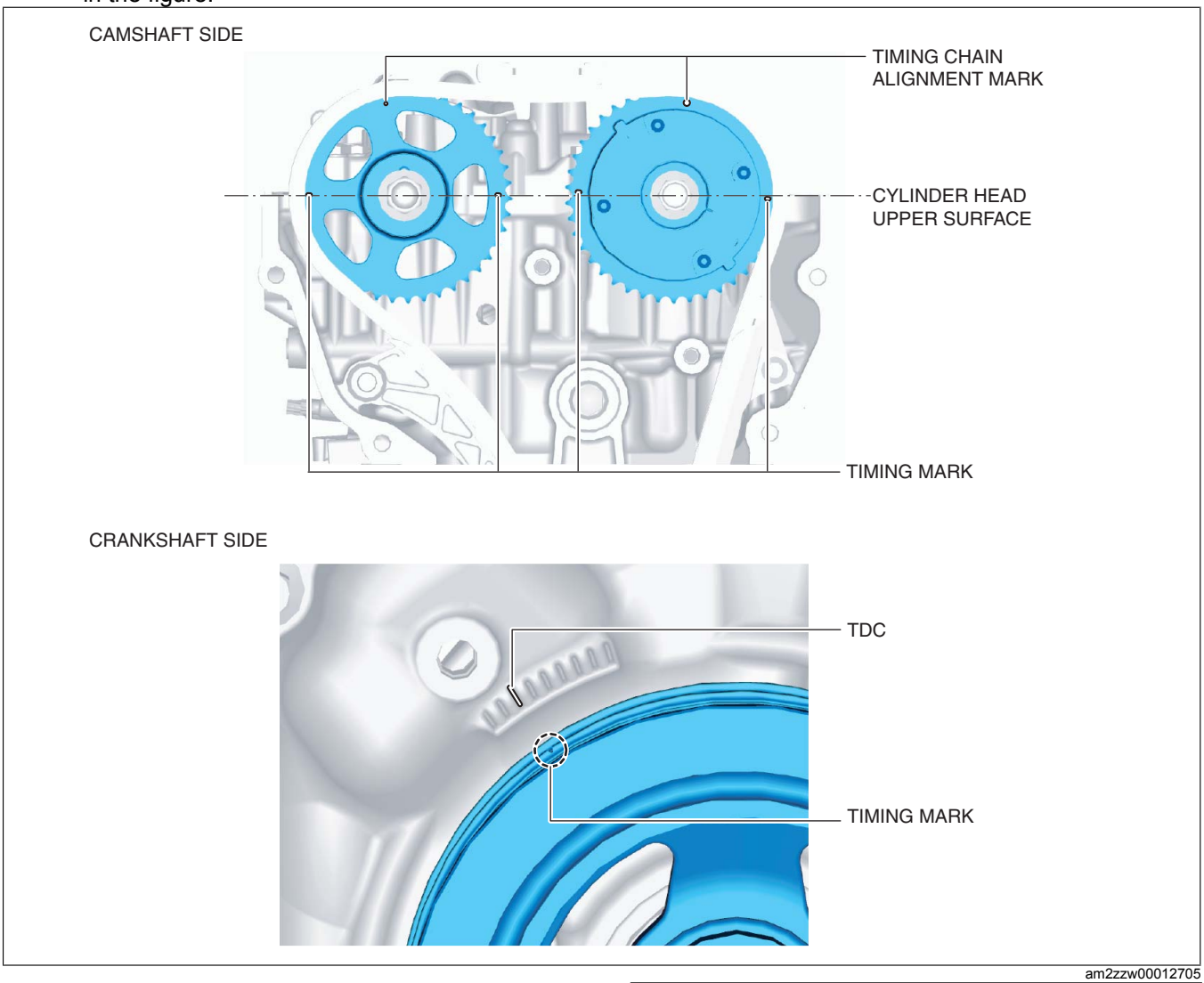
Timing chain removal note

1. Release the tension on the timing chain using the following procedure:

Note

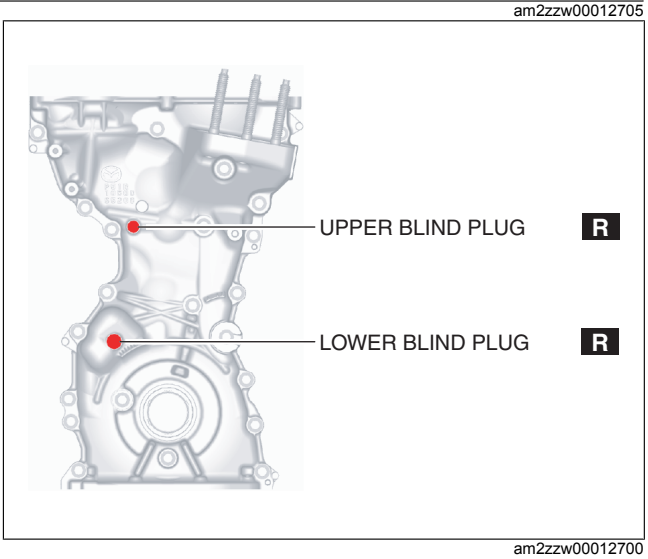
- Release the tension on the timing chain by securing the timing chain tensioner arm through the service hole on the engine front cover.

(1) Rotate the crankshaft clockwise so that cylinder No.1 is positioned around top dead center (TDC) as shown in the figure.



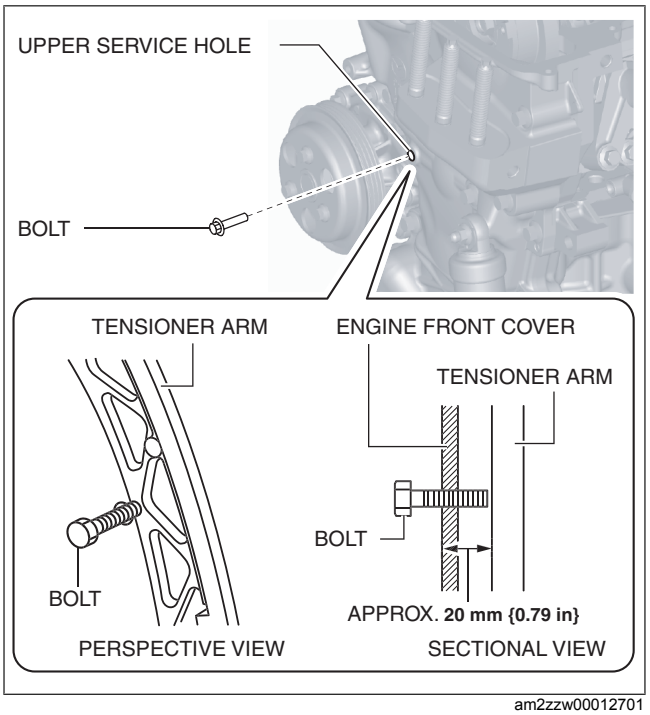
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(2) Remove the blind plugs (upper and lower) on the engine front cover service holes.



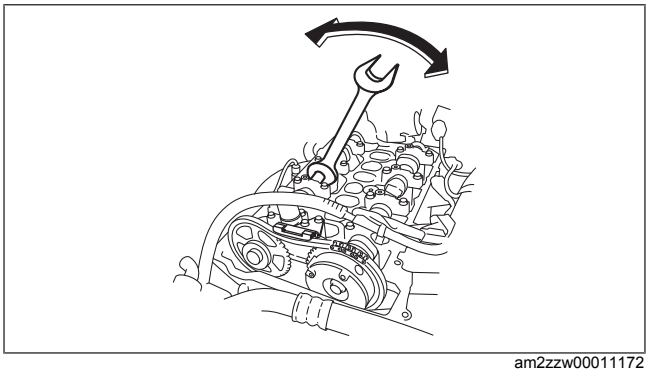
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(3) Insert an **M6** bolt (**35—60 mm {1.4—2.3 in}** length with thread to the end) to the upper service hole and tighten it until it contacts the tensioner arm, then loosen it **approx. 180 degrees** (set the bolt so that it sits slightly in front of the tensioner arm).



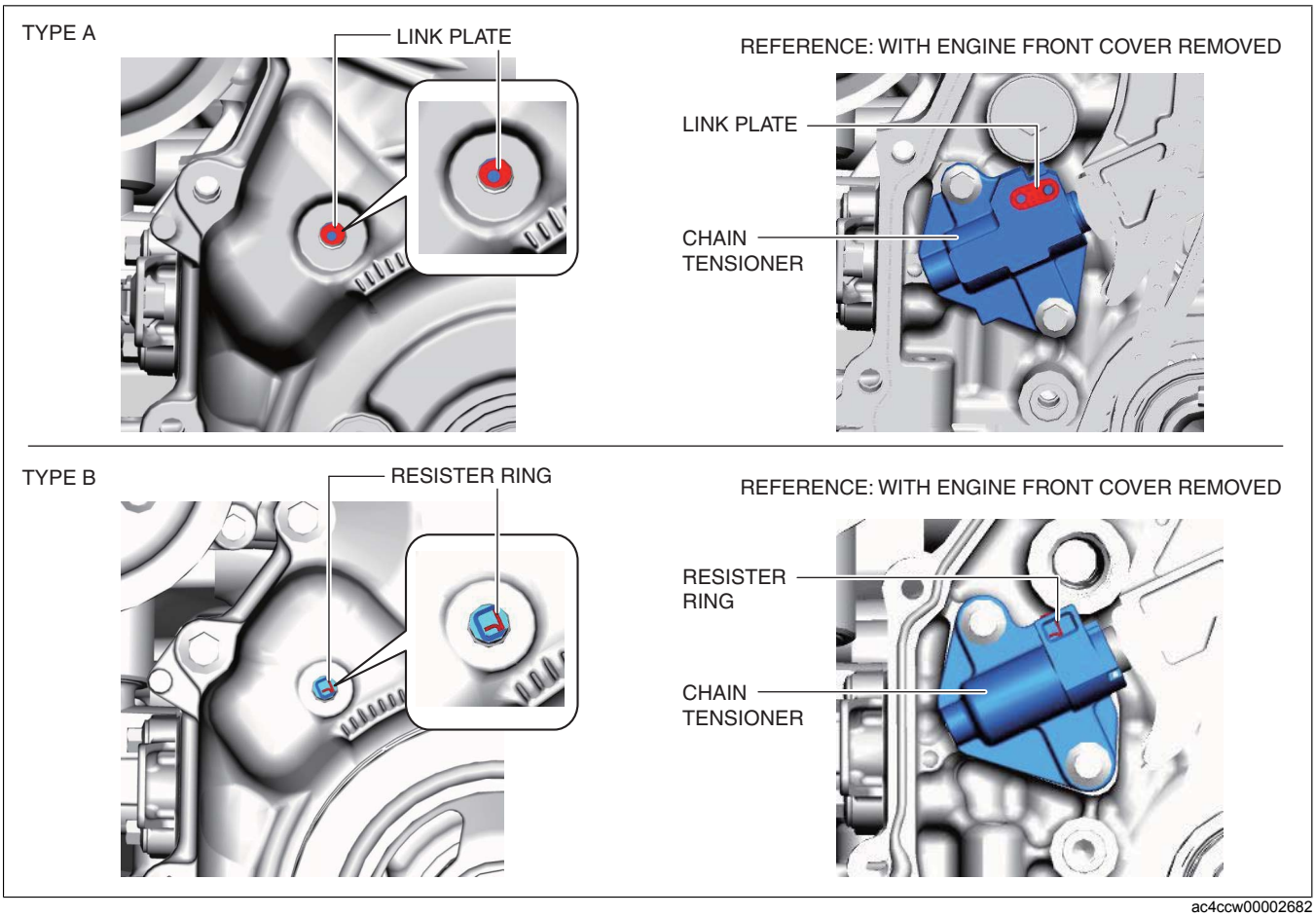
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Note
- The bolt contacts the tensioner arm when it is inserted **approx. 20 mm {0.79 in}**.
- The edge of the bolt in the upper service hole can be verified from the engine upper side (gap between engine front cover and engine).
- Hold the exhaust camshaft using a wrench on the cast hexagon and move it back and forth a few times as shown in the figure. This will drain the oil from the chain tensioner and make the following servicing easier.


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(4) Verify the chain tensioner shape from the lower service hole and identify the chain tensioner type.

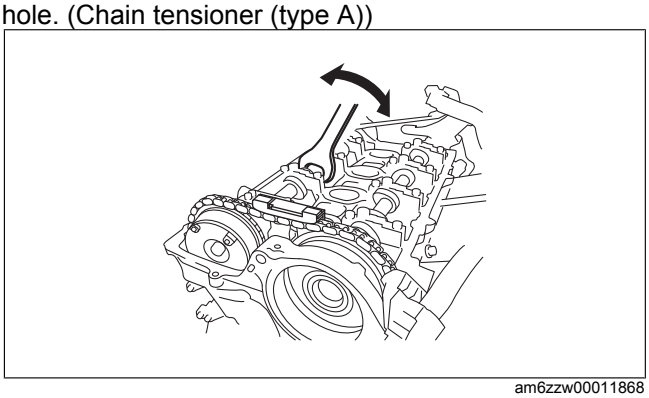
Note
- Verifying the chain tensioner type is necessary because the following procedures may differ according to the type of tensioner.



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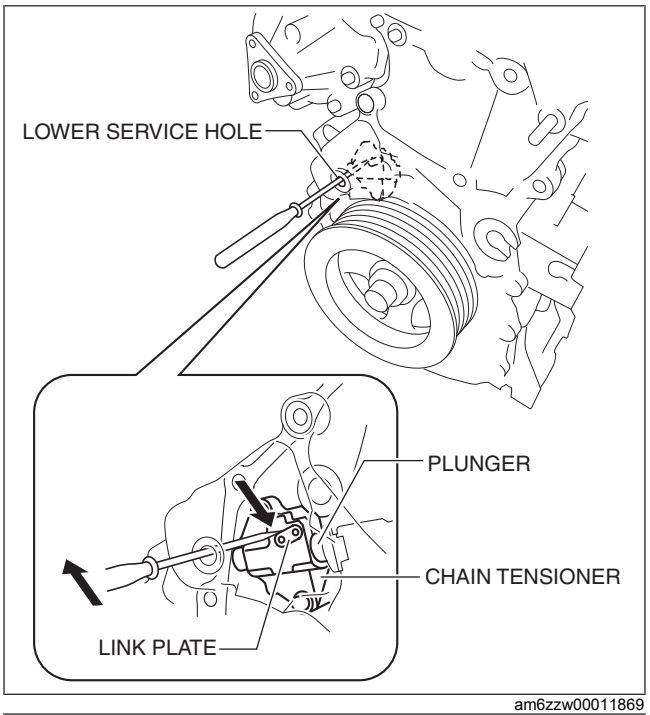
(5) Insert a precision screwdriver into the lower service hole. (Chain tensioner (type A))

(6) While moving the exhaust camshaft back and forth in the direction of the arrow using a wrench on the cast hexagon, press down the link plate of the timing chain tensioner using a precision screwdriver and release the plunger lock. (Chain tensioner (type A))



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Note
- When moving the exhaust camshaft back and forth, the timing chain pushes the plunger in the chain tensioner making it easier to operate the link plate.

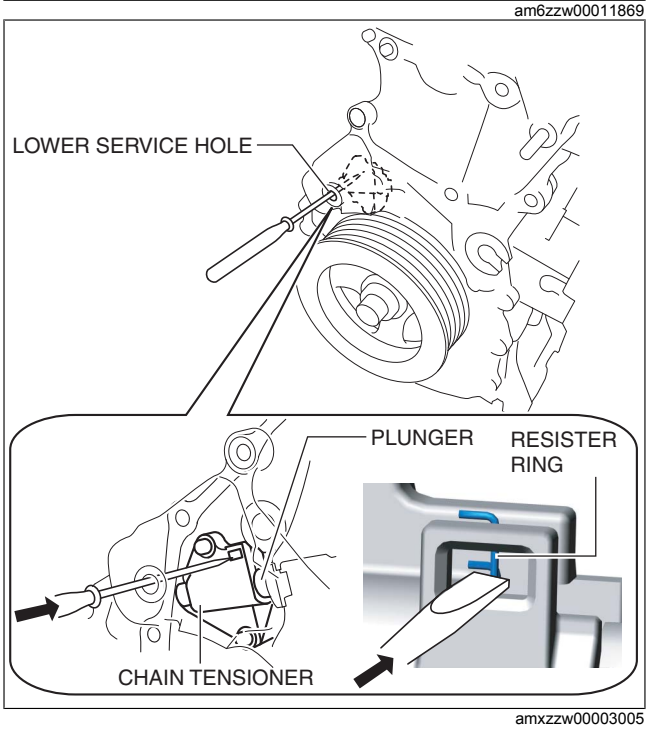


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(7) Insert a thin flathead screwdriver (precision screwdriver) into the lower service hole and press the chain tensioner resister ring. (Chain tensioner (type B))

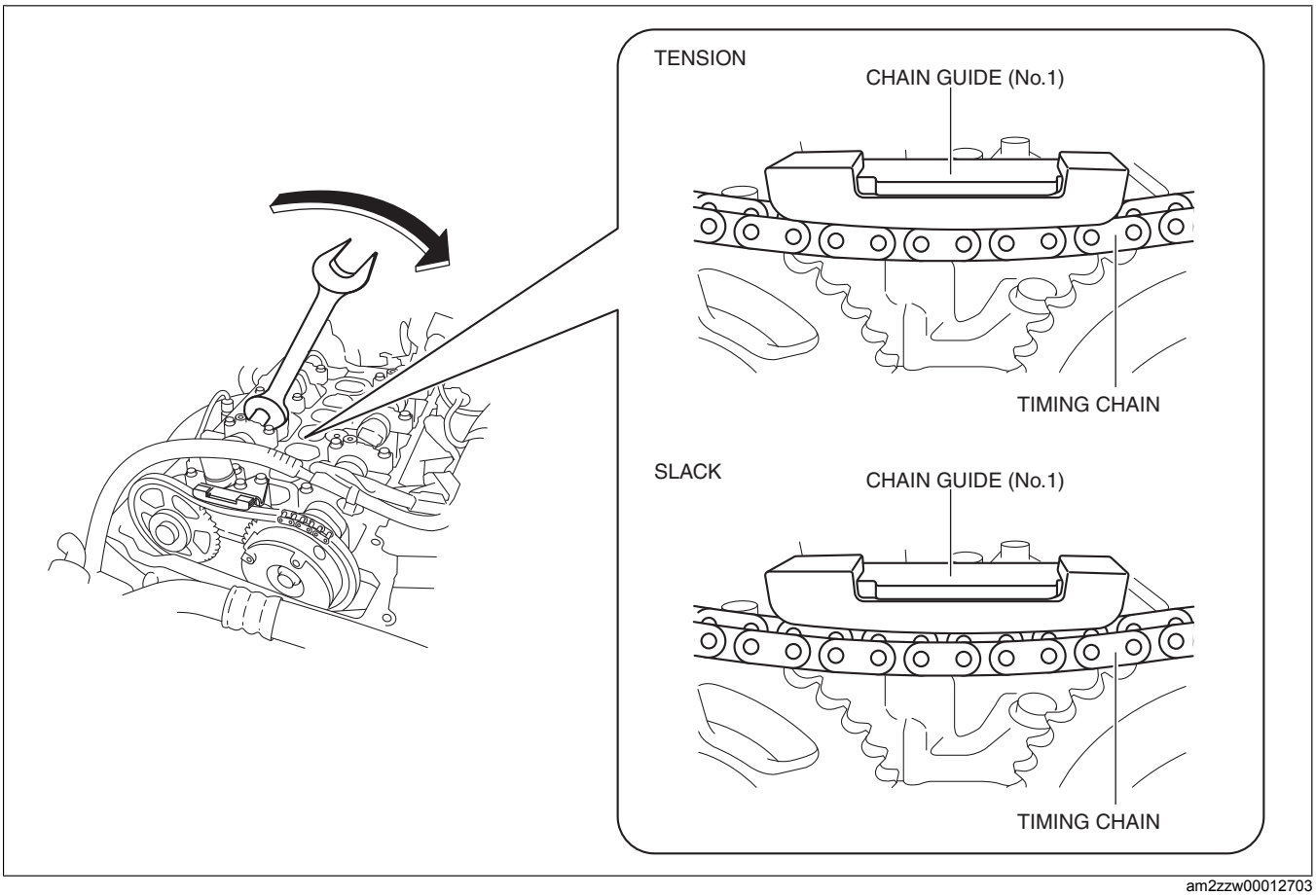
Caution
- **If excessive force is applied to the resister ring, it may cause the resister ring to deform. When pressing the resister ring, be careful not to apply excessive force.**

Note
- The plunger lock is released by pressing the chain tensioner resister ring.



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(8) With the plunger lock released, rotate the exhaust camshaft clockwise until the timing chain loosens.

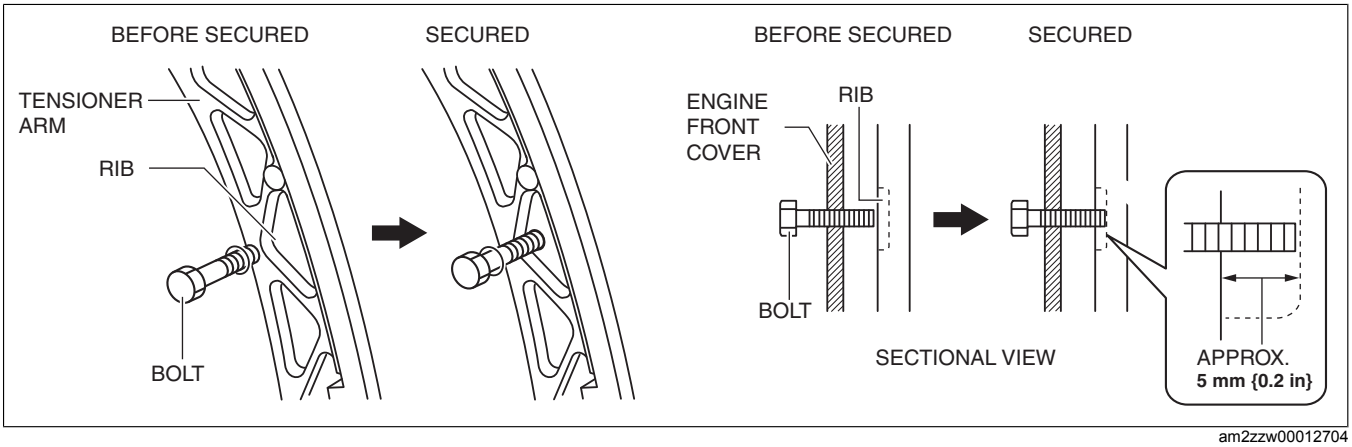


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(9) With the chain loosened, tighten the **M6** bolt in the upper service hole so that the bolt is inserted **approx. 5 mm {0.2 in}** additionally to secure the tensioner arm.

Note

- By rotating the exhaust camshaft clockwise, the timing chain pushes the tensioner arm and the rib position deviates. The entire tensioner arm can be secured by hooking the bolt onto the deviated rib.



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- If the bolt cannot be inserted **approx. 5 mm {0.2 in}**, the plunger lock of the chain tensioner may not release, or the chain may not loosen enough. Return the bolt to the original position and restart the procedure from Step (3).

(10) After securing the tensioner arm, remove the chain guide (No.1).

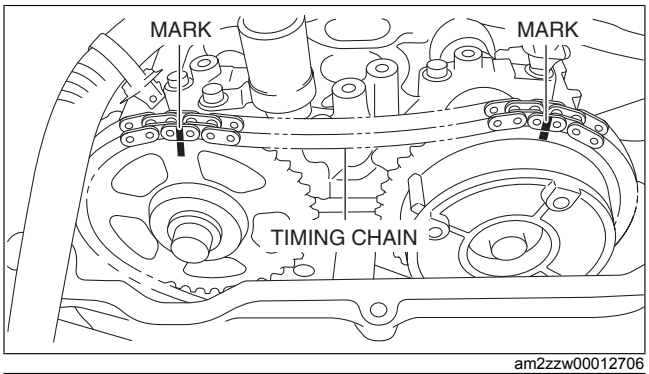
Exhaust camshaft sprocket removal note

Caution

- **Keep the timing chain pulled up during and after the work in which it has been set aside from the sprocket. If the chain falls down, it may disengage from the sprocket on the crankshaft side and deviate, leading to valve timing deviation.**

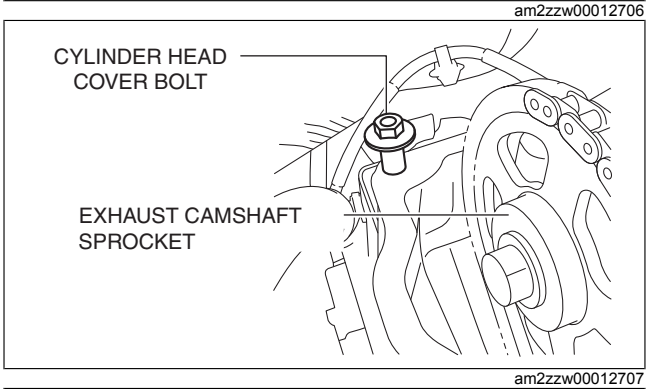
- **After servicing, always verify the valve timing.**

1. Mark the timing chain, hydraulic variable valve timing actuator, and exhaust camshaft sprocket so that they can be reassembled in the same condition as before removal.



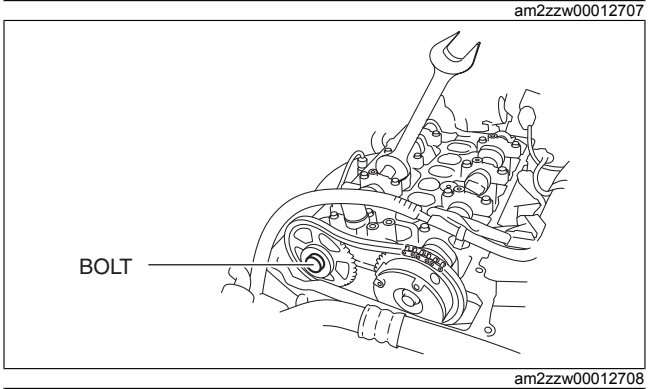
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2. Temporarily install the cylinder head cover bolt to the position shown in the figure to use it for suspending the timing chain.



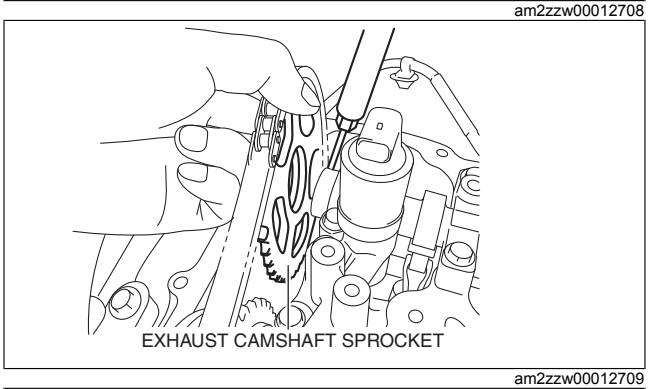
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3. Hold the exhaust camshaft using a wrench on the cast hexagon, and loosen the exhaust camshaft sprocket installation bolt.
4. Remove the exhaust camshaft sprocket installation bolt.



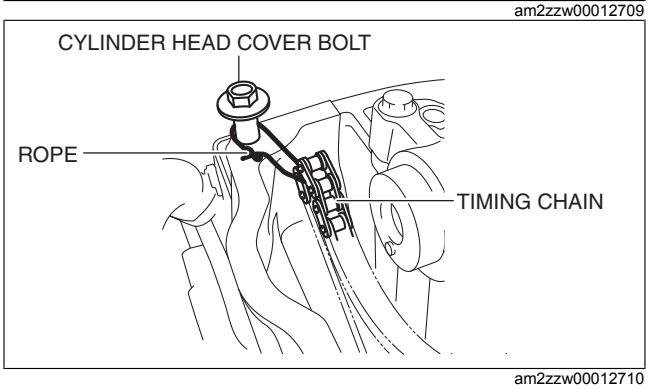
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5. Using a screwdriver wrapped in a cloth, lightly press the exhaust camshaft sprocket to the front and separate the exhaust camshaft and the sprocket.
6. Remove the exhaust camshaft sprocket.



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7. Pull up the timing chain and suspend it using a rope as shown in the figure.



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Camshaft removal note

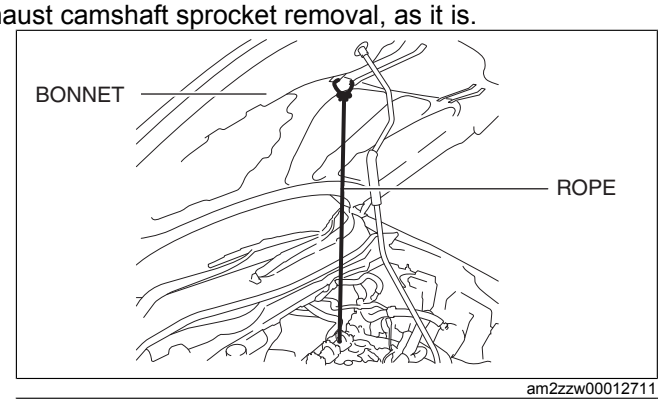
Caution

- **Keep the timing chain pulled up during and after the work in which it has been set aside from the sprocket. If the chain falls down, it may disengage from the sprocket on the crankshaft side and deviate, leading to valve timing deviation.**
- **After servicing, always verify the valve timing.**

Note

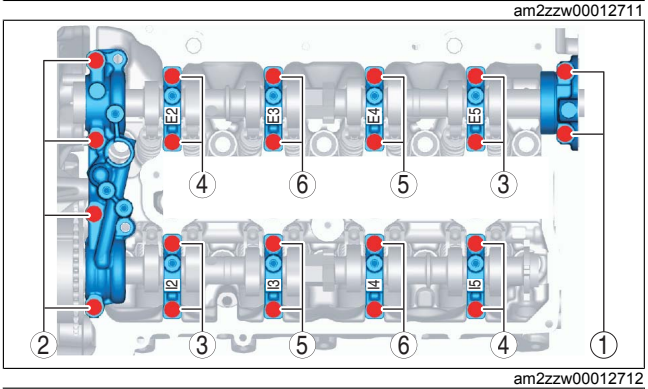
- Remove the intake camshaft and the hydraulic variable valve timing actuator as a single unit.

1. Leave the timing chain which was suspended after exhaust camshaft sprocket removal, as it is.
2. Suspend a string from the hood to use for suspending the timing chain.



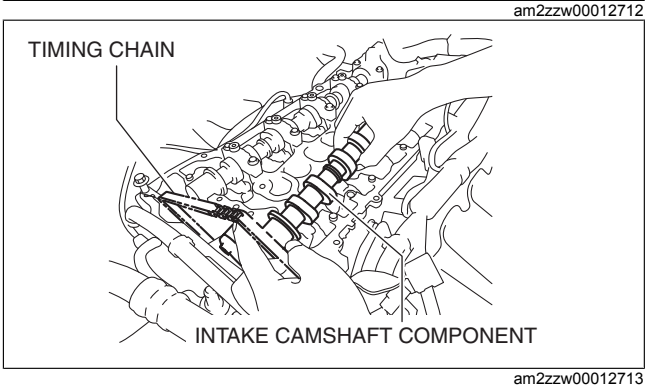
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3. Loosen the camshaft cap installation bolts in a few passes in the order shown in the figure and remove the camshaft caps.



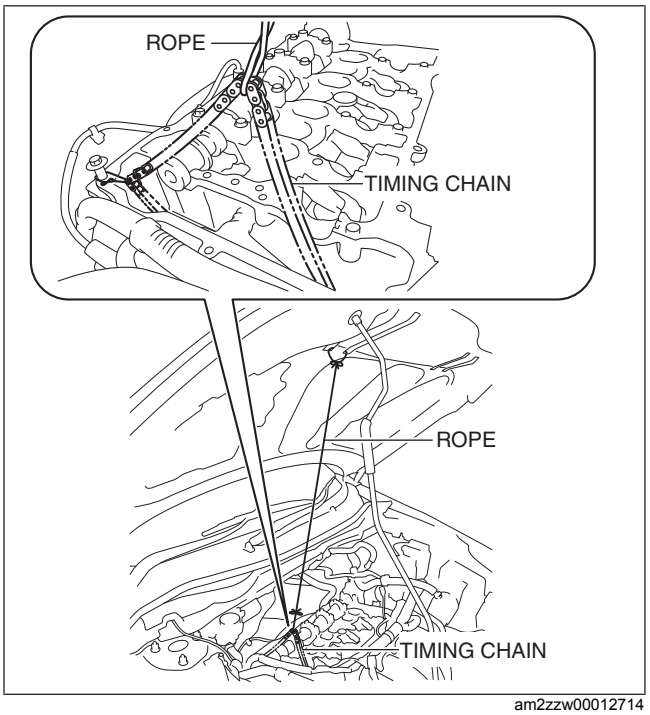
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4. While setting the timing chain aside, remove the hydraulic variable valve timing actuator and the intake camshaft as a single unit.



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5. Pull up the timing chain and suspend it using a rope as shown in the figure.
6. Remove the exhaust camshaft.



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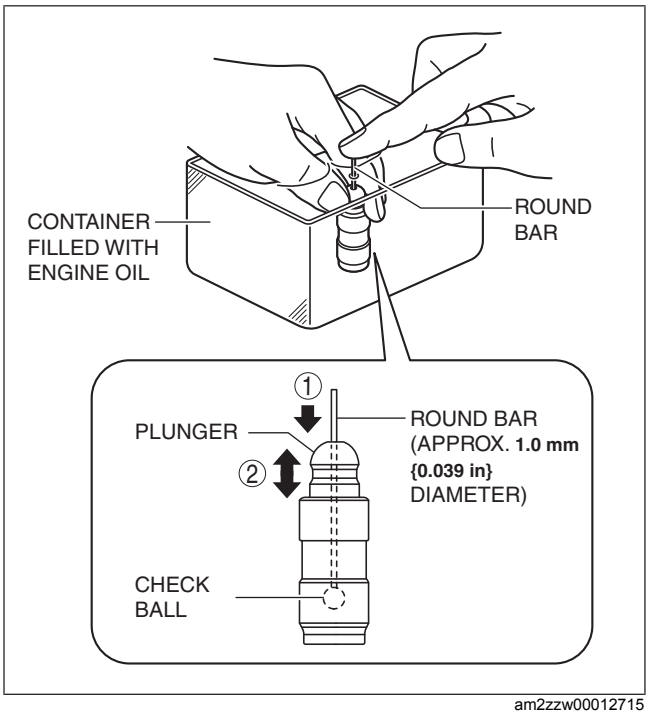
Rocker arm removal note
1. Keep the rocker arms in the order of removal to enable reassembly in their original positions.

HLA removal note
1. Keep the HLAs in the order of removal to enable reassembly in their original positions.

HLA installation note
1. Perform HLA air bleeding using the following procedure.
   (1) Put the HLA in a container filled with engine oil.

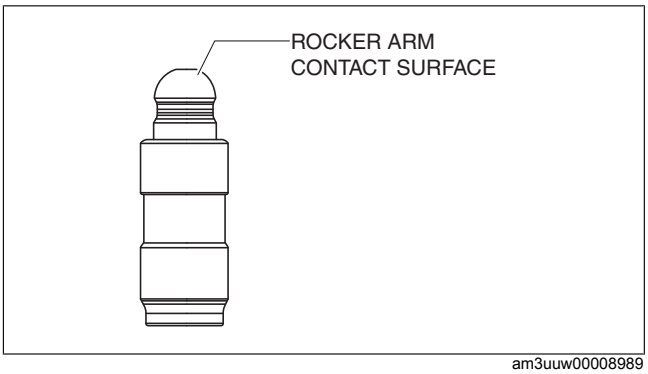
Caution
- **Do not insert the round bar firmly because the check ball spring force is extremely weak.**

   (2) While lightly pressing the check ball using a round bar (**approx. 1.0 mm {0.039 in}** diameter), bleed air by moving the plunger up and down.
   (3) Press the end of the plunger in the oil and verify that there is no rebounding feel.
      - If rebounding feel cannot be eliminated, replace the HLA.



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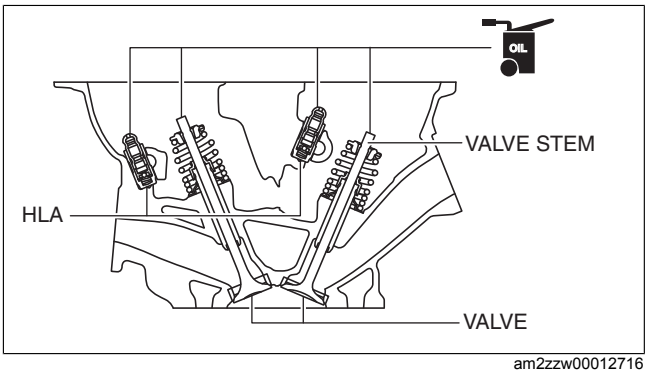
2. Visually inspect the HLA surface where it contacts the rocker arm for wear or damage.
   - If there is any malfunction, replace the HLA.
3. Assemble the HLAs to their original positions.



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Rocker arm installation note

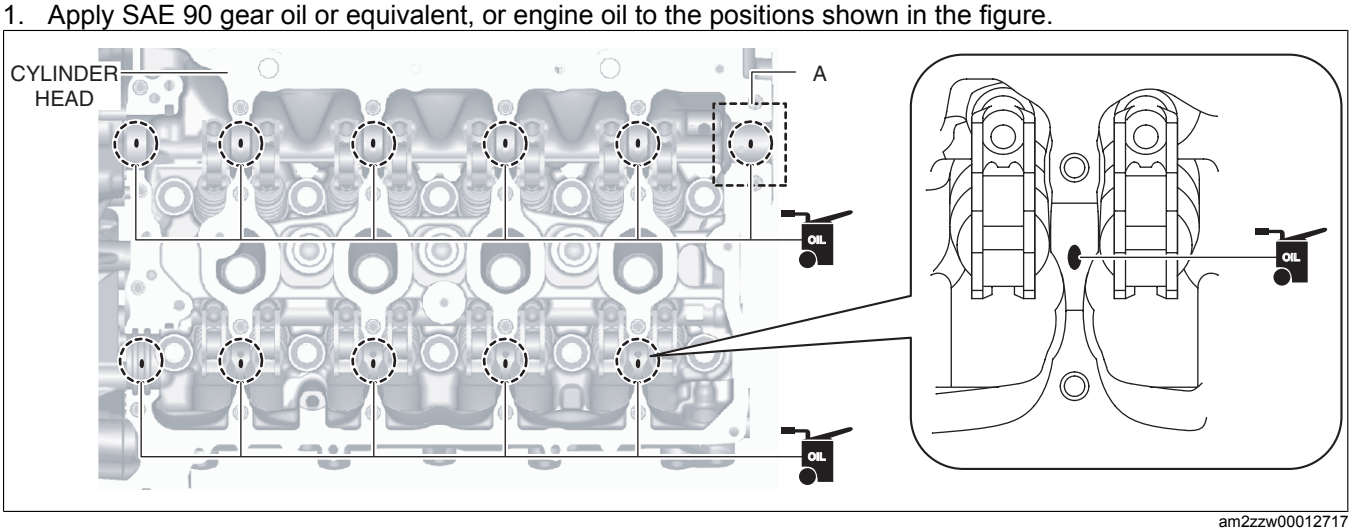
1. Apply engine oil to the HLAs and the end of the valve stems.
2. Install the rocker arms to the same positions as before removal.



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Camshaft installation note

1. Apply SAE 90 gear oil or equivalent, or engine oil to the positions shown in the figure.



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Caution

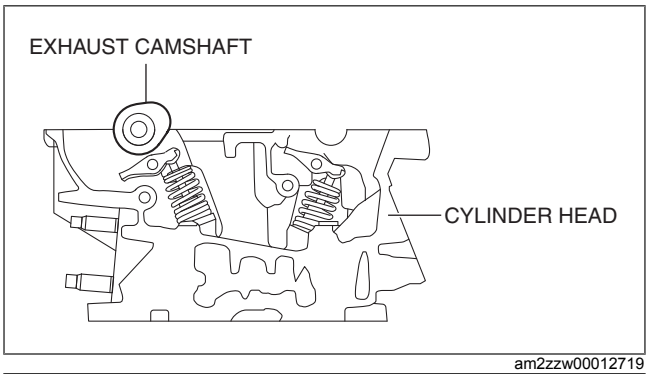
- **Apply 0.05 ml {0.05 cc, 0.003 in^3} or less of oil to area A in the figure.**

2. Apply SAE 90 gear oil or equivalent, or engine oil to the following locations of each camshaft.
   - Thrust surface of front journal (both surfaces front and back)

Note

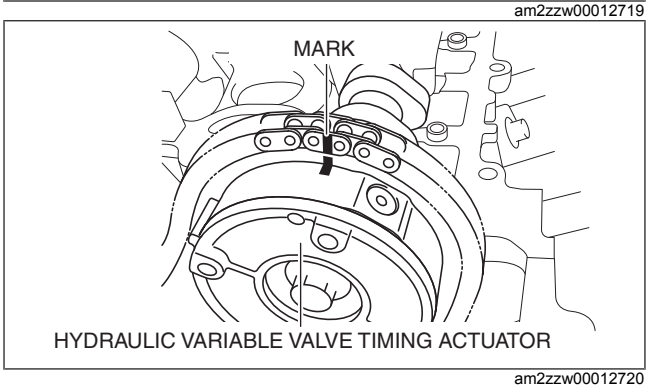
- If oil is applied to the front camshaft cap, oil should not be applied to the thrust surface of the front journal.

3. Align the cam position of cylinder No.1 around top dead center (TDC) and place the exhaust camshaft on the cylinder head.
4. Remove the rope suspending the timing chain on the intake side.



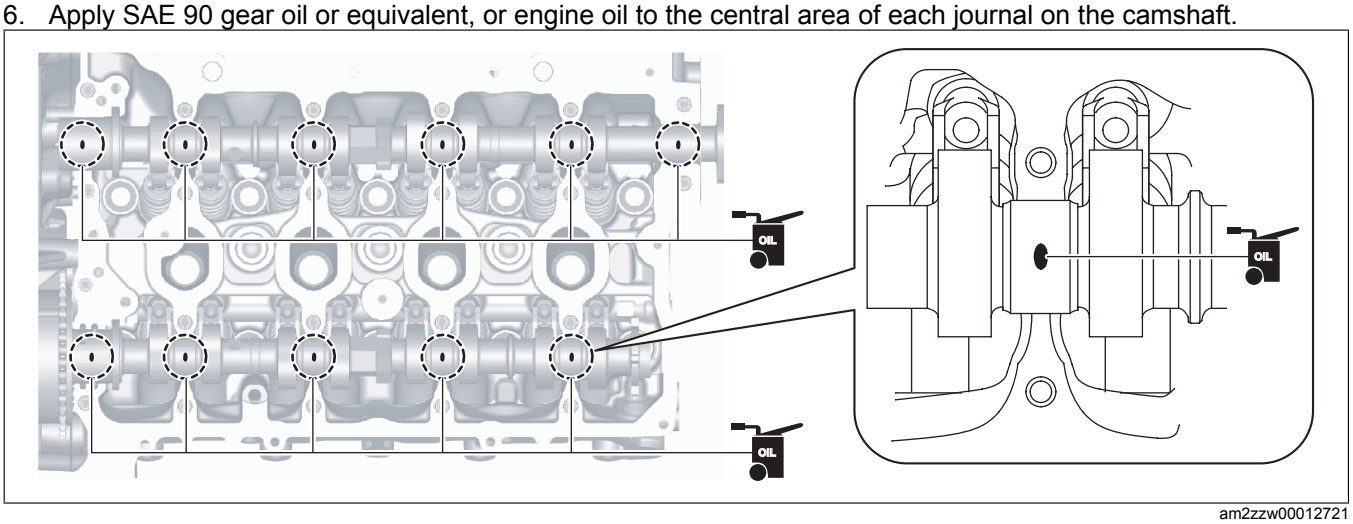
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5. While aligning the mark on the timing chain placed during removal with the one on the hydraulic variable valve timing actuator, place the intake camshaft component on the cylinder head.



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6. Apply SAE 90 gear oil or equivalent, or engine oil to the central area of each journal on the camshaft.



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7. Apply SAE 90 gear oil or equivalent, or engine oil to the thrust surface of the front camshaft cap.

Note
- If oil is applied to the front journal thrust surface of each camshaft, oil should not be applied to the front camshaft cap.

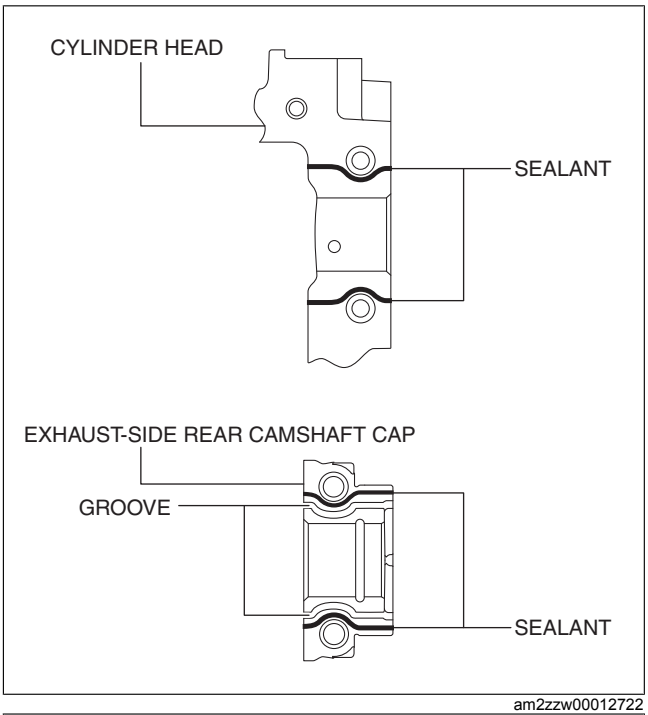
8. Apply sealant (Loctite #962T or equivalent) to the rear camshaft cap installation area on the exhaust side or the rear camshaft caps on the exhaust side of the cylinder head.

Note
- To prevent engine oil leakage, apply sealant to the rear camshaft cap installation area on the exhaust side or the rear camshaft caps on the exhaust side of the cylinder head, and seal the journal.

Caution
- **Do not spill sealant on the journal.**

Sealant bead width
0.5—1.5 mm {0.02—0.05 in}

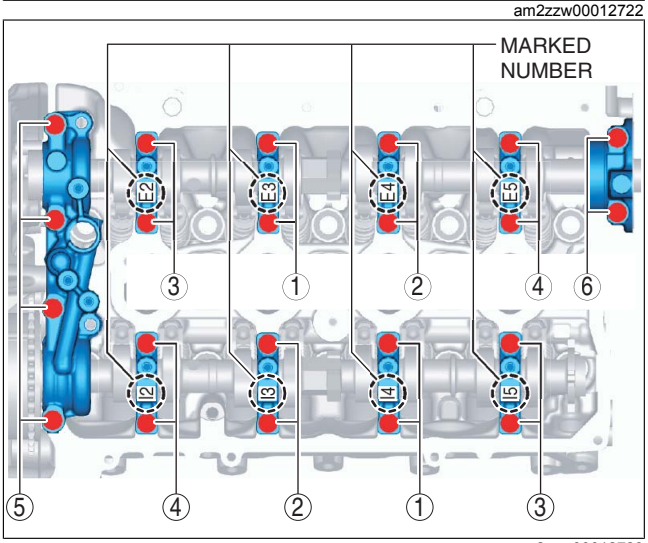
9. Install the camshaft caps in the order of the marked numbers, and temporarily tighten the camshaft cap installation bolts evenly in **2—3 rounds**.



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10. Tighten the camshaft cap installation bolts in two steps in the order shown in the figure.

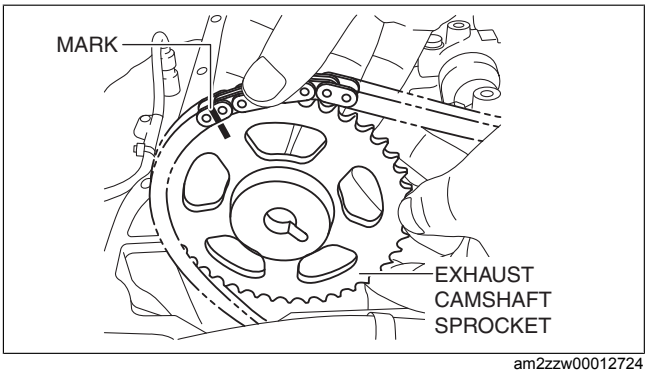
Tightening torque
Step 1: 3.0—6.0 N·m {31—61 kgf·cm, 27—53 in·lbf}
Step 2: 8—11 N·m {82—112 kgf·cm, 71—97 in·lbf}



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Exhaust camshaft sprocket installation note

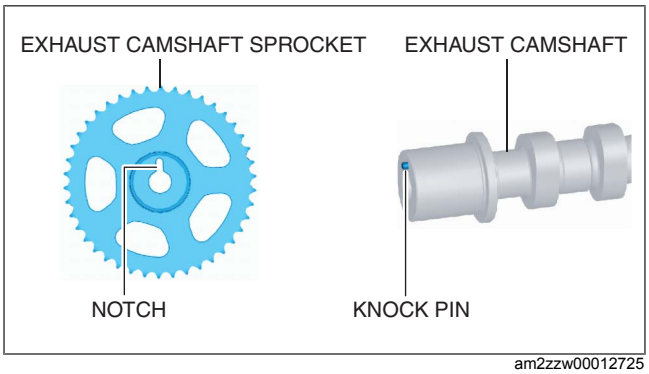
1. Remove the rope suspending the timing chain on the exhaust side.
2. While aligning the mark on the timing chain marked during removal with the one on the exhaust camshaft sprocket, engage the chain with the sprocket.



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3. Align the knock pin on the end of the exhaust camshaft with the notch on the exhaust camshaft sprocket side, then assemble the sprocket.

Note
- Adjust the knock pin with the notch while adjusting the knock pin position by rotating the camshaft.

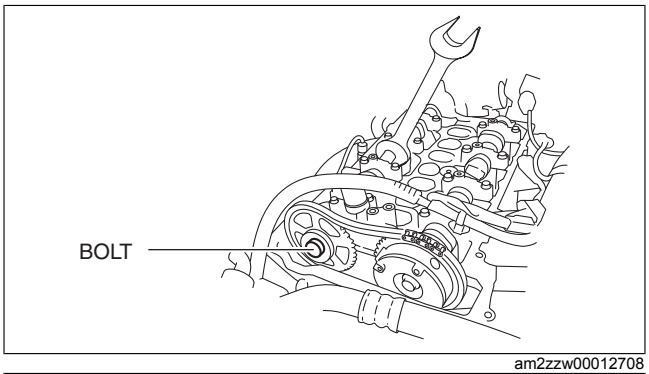


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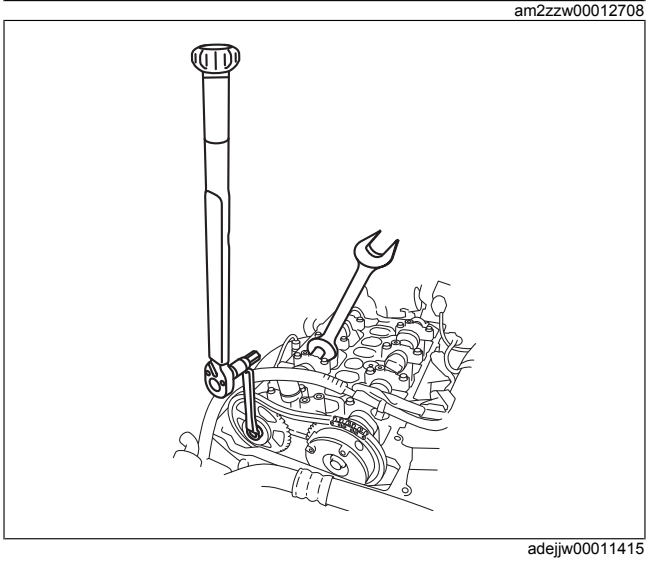
4. Tighten the exhaust camshaft sprocket installation bolt using the following procedure:
   (1) Temporarily tighten the exhaust camshaft sprocket installation bolt.
   (2) Hold the exhaust camshaft using a wrench on the cast hexagon, and tighten the exhaust camshaft sprocket installation bolt.

Tightening torque
123—140 N·m {13—14 kgf·m, 91—103 ft·lbf}

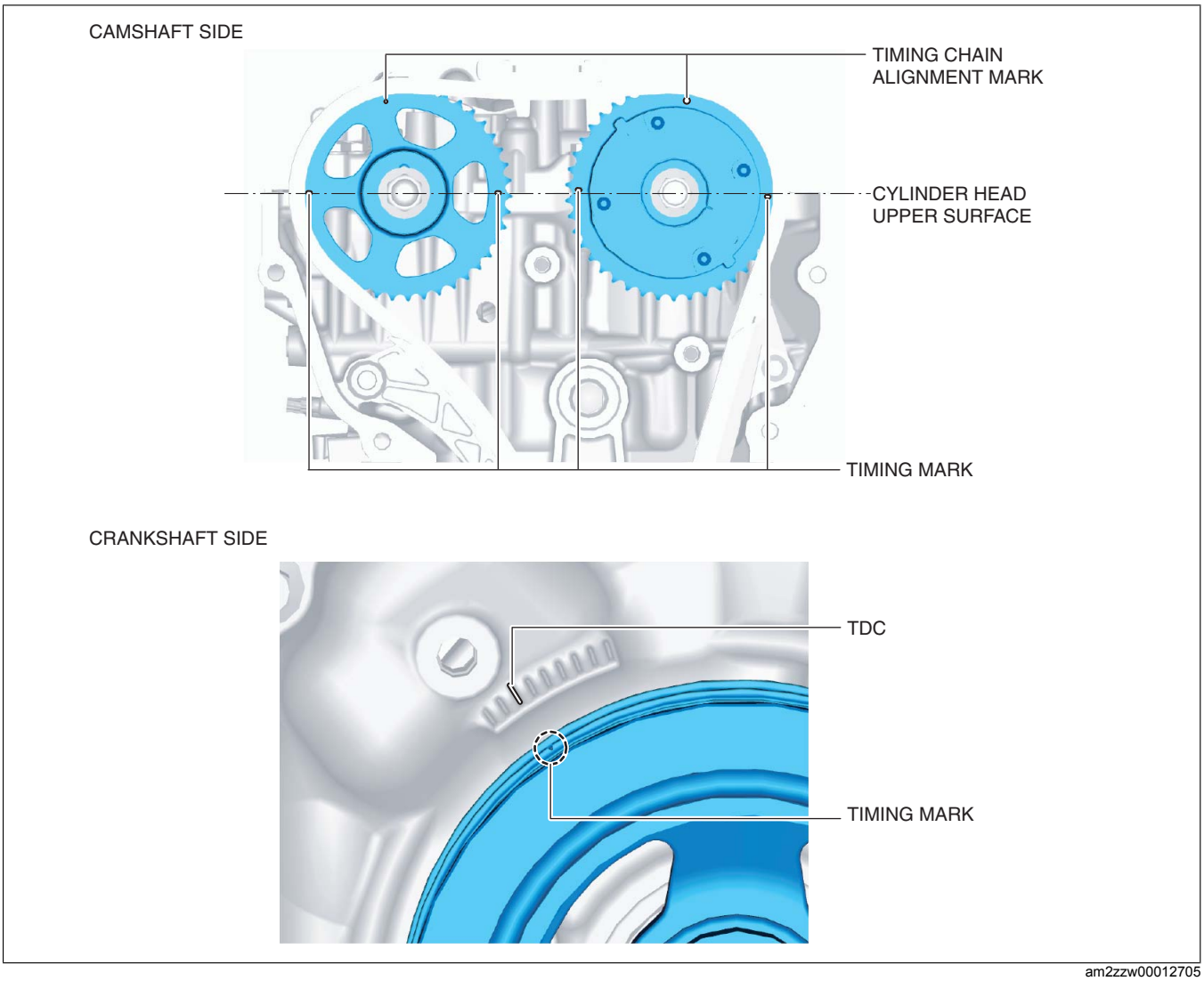
Note
- If there is no appropriate tool which can be used to tighten the camshaft sprocket installation bolt to the specified tightening torque, tighten the bolt using the method as shown in the figure. For the specified tightening torque calculation method if the tool is elongated, refer to the [SERVICE CAUTIONS] in the general information. (See SERVICE CAUTIONS.)



am2zzw00012708


adejjw00011415

Timing chain installation note
1. Install the chain guide (No.1).
2. Remove the bolt securing the tensioner arm and apply tension to the timing chain.
3. Rotate the crankshaft clockwise **two turns** and verify that the valve timing is correct.



am2zzw00012705

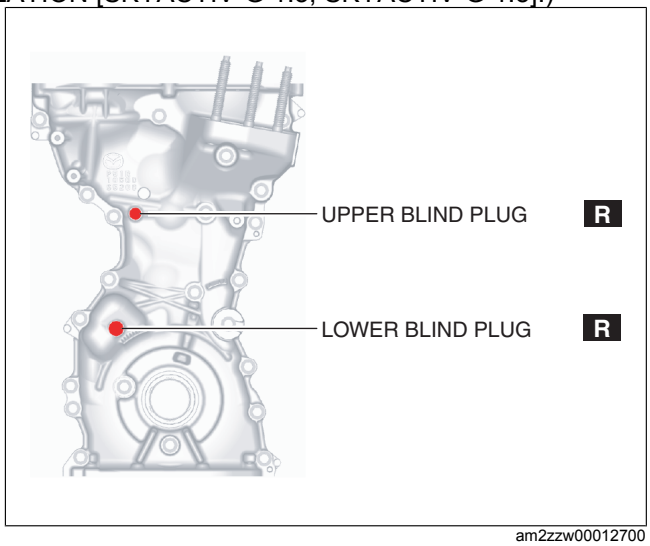
Note

- The timing mark is not completely parallel with the upper surface of the cylinder head.

- If the valve timing is not correct, remove the engine front cover and reassemble the timing chain to each sprocket. (See TIMING CHAIN REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)

4. Install the blind plugs (upper and lower) for the engine front cover service holes.

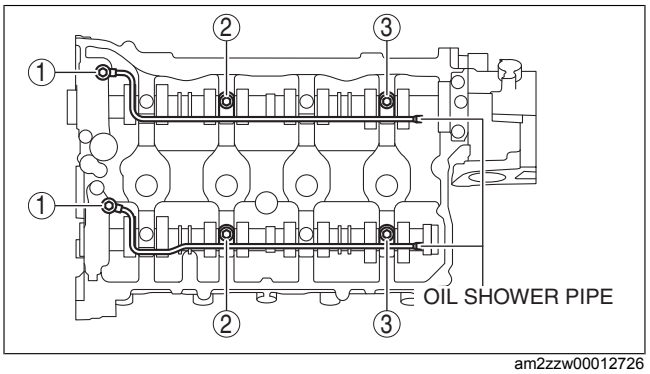
Tightening torque
Upper: 8—11 N·m {82—112 kgf·cm, 71—97 in·lbf}
Lower: 17—23 N·m {1.8—2.3 kgf·m, 13—16 ft·lbf}



am2zzw00012700

Oil shower pipe installation note

1. Install the oil shower pipe in the order shown in the figure.



am2zzw00012726

Tightening torque

Installation position	Tightening torque
1	9—12 N·m {92—122 kgf·cm, 80—106 in·lbf}
2, 3	8.0—9.0 N·m {82—91 kgf·cm, 71—79 in·lbf}

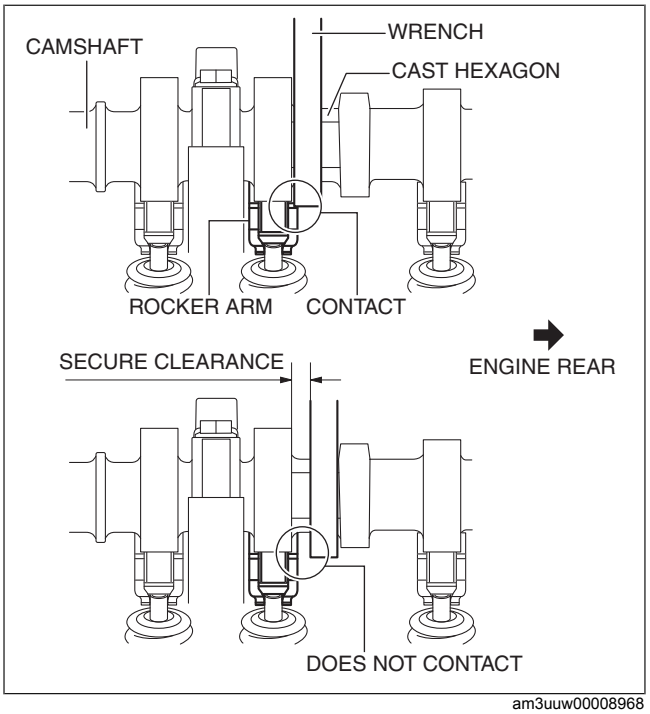
SKYACTIV-G 1.5 (with 4-2-1 exhaust system)

Warning

- **A hot engine can cause severe burns. Turn off the engine and wait until it is cool before servicing.**

Caution

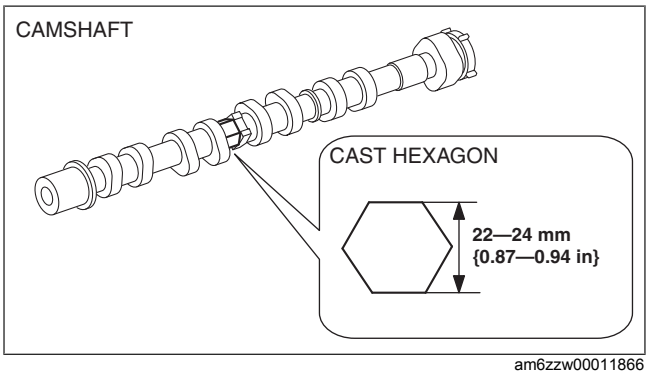
- **If the camshaft is rotated with the timing chain removed and the piston at the top dead center position, the valve may contact the piston and the engine could be damaged. When rotating the camshaft with the timing chain removed, rotate it after lowering the piston from the top dead center position.**
- **When rotating the camshaft using a wrench on the cast hexagon, the wrench may contact the rocker arm and damage the rocker arm. To prevent damage to the rocker arm when holding the camshaft on the cast hexagon, use a wrench on the rear side of the engine as shown in the figure to secure a clearance between the cam.**



am3uuw00008968

Note

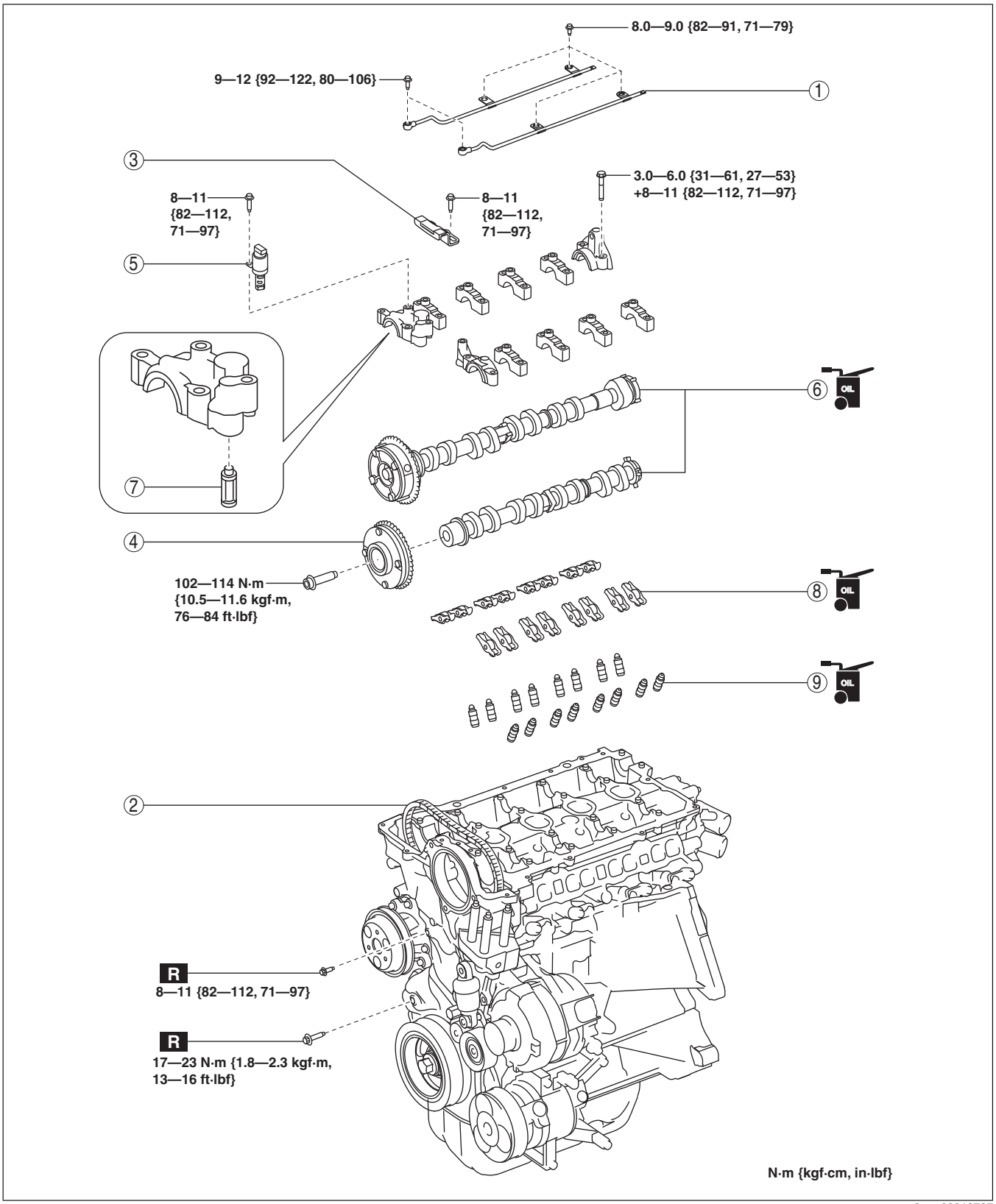
- Width at the cast hexagon of the camshaft is **22—24 mm {0.87—0.94 in}**.



am6zzw00011866

1. Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.)
2. Remove the plug hole plate. (See PLUG HOLE PLATE REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
3. Remove the ignition coil/ion sensors. (See IGNITION COIL/ION SENSOR REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)

4. Remove the cylinder head cover. (See TIMING CHAIN REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
5. Remove the electric variable valve timing motor/driver. (See ELECTRIC VARIABLE VALVE TIMING MOTOR/DRIVER REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
6. Remove the vacuum pump. (See VACUUM PUMP REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
7. Remove the high pressure fuel pump and rear housing. (See HIGH PRESSURE FUEL PUMP REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
8. Remove in the order indicated in the table.
9. Install in the reverse order of removal.
10. Start the engine and inspect the following:
    - Ignition timing, idle speed and idle mixture. (See ENGINE TUNE-UP [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)



am2zzw00012727

1	Oil shower pipe (See Oil shower pipe installation note.)
2	Timing chain (See Timing chain removal note.) (See Timing chain installation note.)
3	Chain guide (No.1)

4	Electric variable valve timing actuator (See Electric variable valve timing actuator removal note.) (See Electric variable valve timing actuator installation note.)
5	OCV
6	Camshaft (See Camshaft removal note.) (See Camshaft installation note.)

7	OCV oil filter
8	Rocker arm (See Rocker arm removal note.) (See Rocker arm installation note.)

9	HLA (See HLA removal note.) (See HLA installation note.)

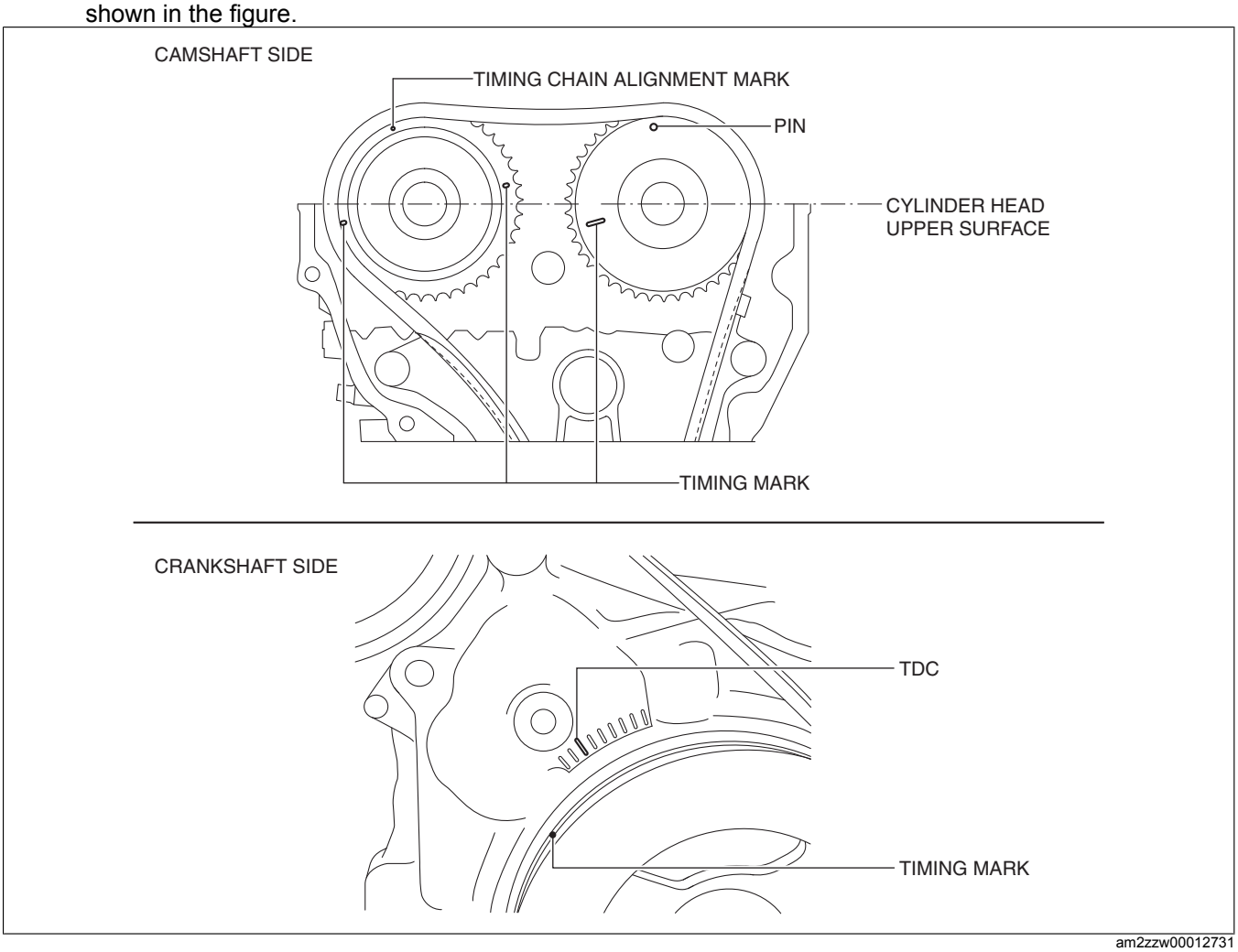
Timing chain removal note

1. Release the tension on the timing chain using the following procedure:

Note

- Release the tension on the timing chain by securing the timing chain tensioner arm through the service hole on the engine front cover.

(1) Rotate the crankshaft clockwise so that cylinder No.1 is positioned around before top dead center (TDC) as shown in the figure.

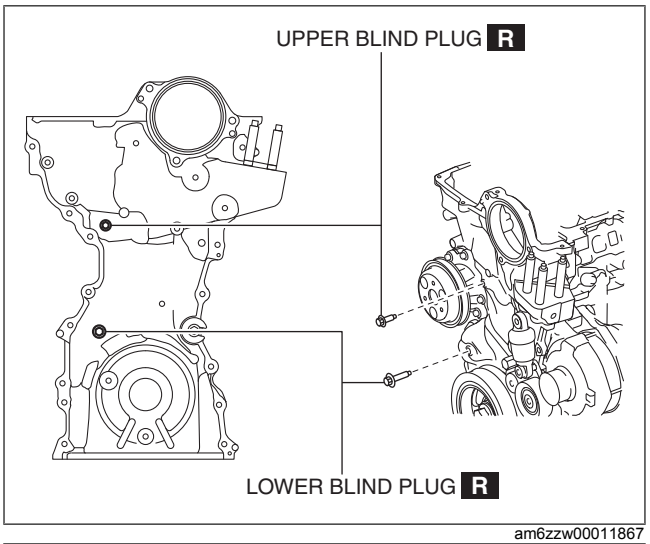


am2zzw00012731

Note

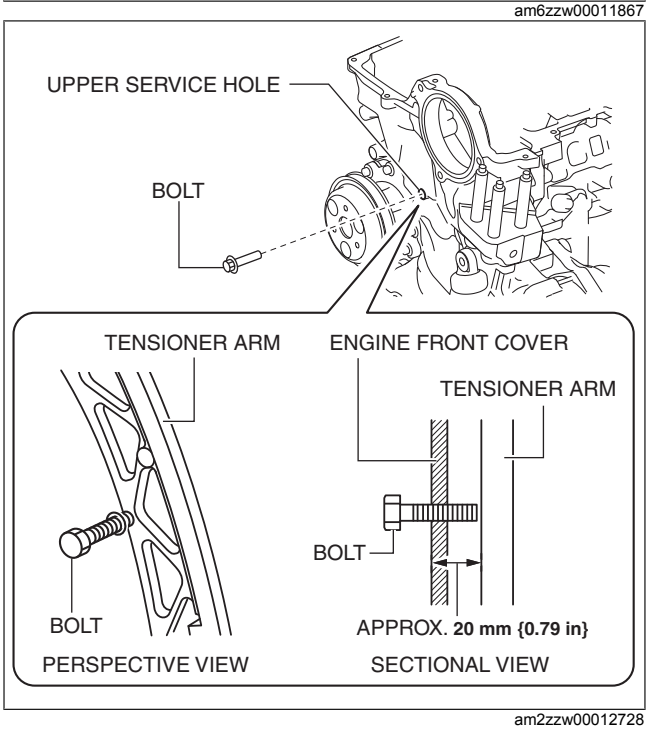
- With cylinder No.1 positioned around before top dead center (TDC), there is no large movement (rotation) of the exhaust camshaft when the timing chain is removed. Therefore, installation of the electric variable valve timing actuator to the camshaft will be easier. (If there is a large movement of the exhaust camshaft, installation of the electric variable valve timing actuator to the camshaft must be performed while loosening the chain by rotating the exhaust camshaft.)

(2) Remove the blind plugs (upper and lower) on the engine front cover service holes.



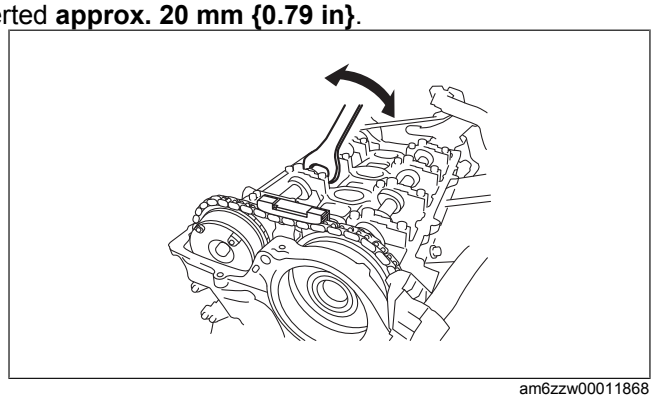
am6zzw00011867

(3) Insert an M6 bolt (**35—60 mm {1.4—2.3 in}** length with thread to the end) to the upper service hole and tighten it until it contacts the tensioner arm, then loosen it **approx. 180 degrees** (set the bolt so that it sits slightly in front of the tensioner arm).



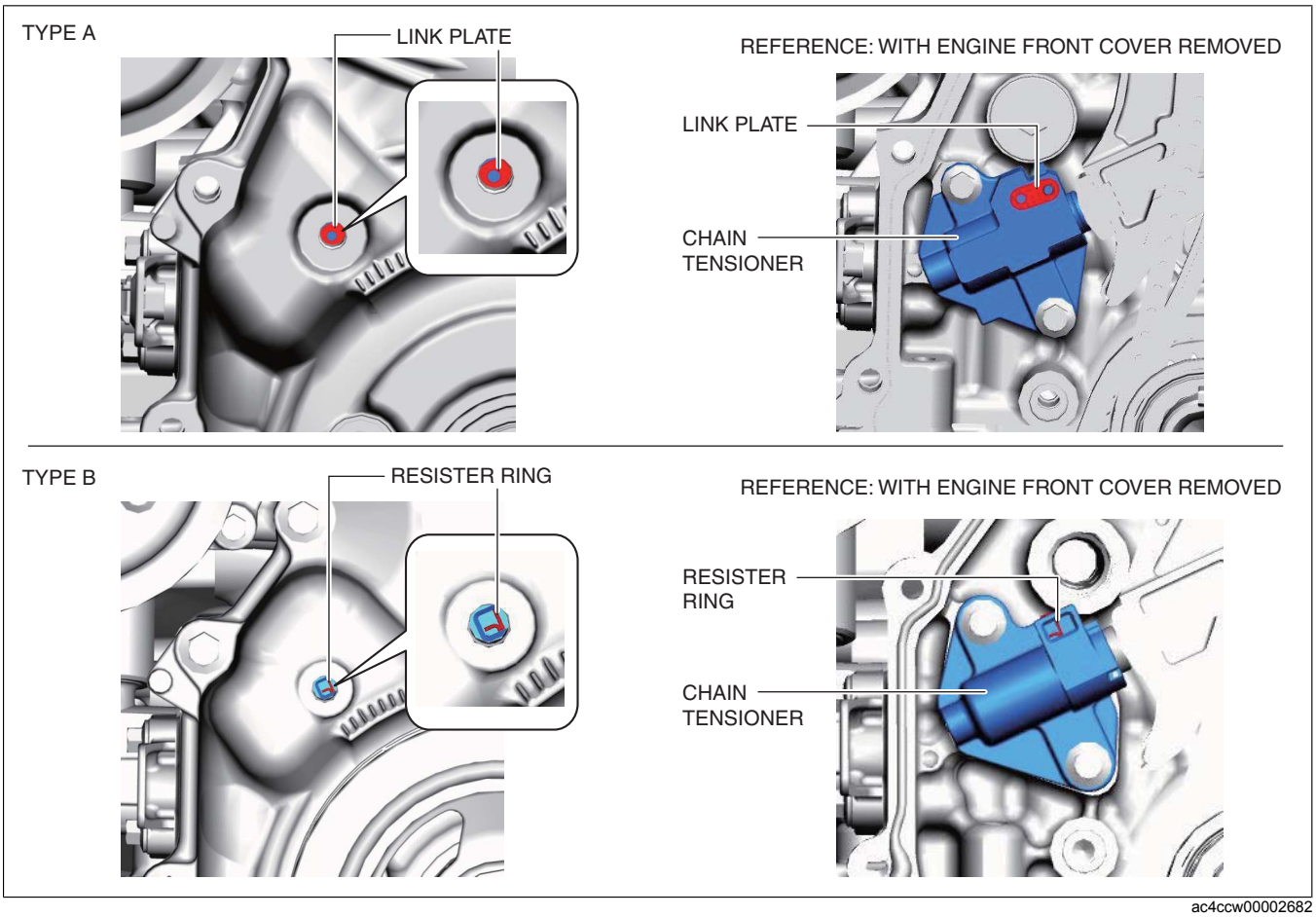
am2zzw00012728

Note
- The bolt contacts the tensioner arm when it is inserted **approx. 20 mm {0.79 in}**.
- Hold the exhaust camshaft using a wrench on the cast hexagon and move it back and forth a few times as shown in the figure. This will drain the oil from the chain tensioner and make the following servicing easier.


am6zzw00011868

(4) Verify the chain tensioner shape from the lower service hole and identify the chain tensioner type.

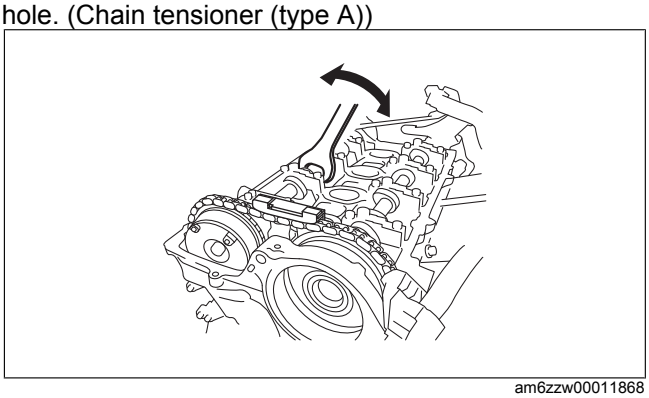
Note
- Verifying the chain tensioner type is necessary because the following procedures may differ according to the type of tensioner.



ac4ccw00002682

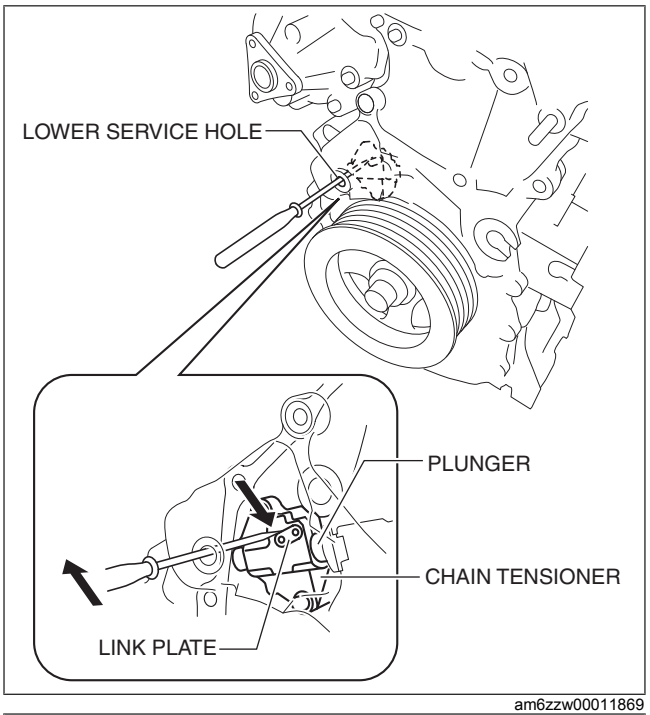
(5) Insert a precision screwdriver into the lower service hole. (Chain tensioner (type A))

(6) While moving the exhaust camshaft back and forth in the direction of the arrow using a wrench on the cast hexagon, press down the link plate of the timing chain tensioner using a precision screwdriver and release the plunger lock. (Chain tensioner (type A))



am6zzw00011868

Note
- When moving the exhaust camshaft back and forth, the timing chain pushes the plunger in the chain tensioner making it easier to operate the link plate.

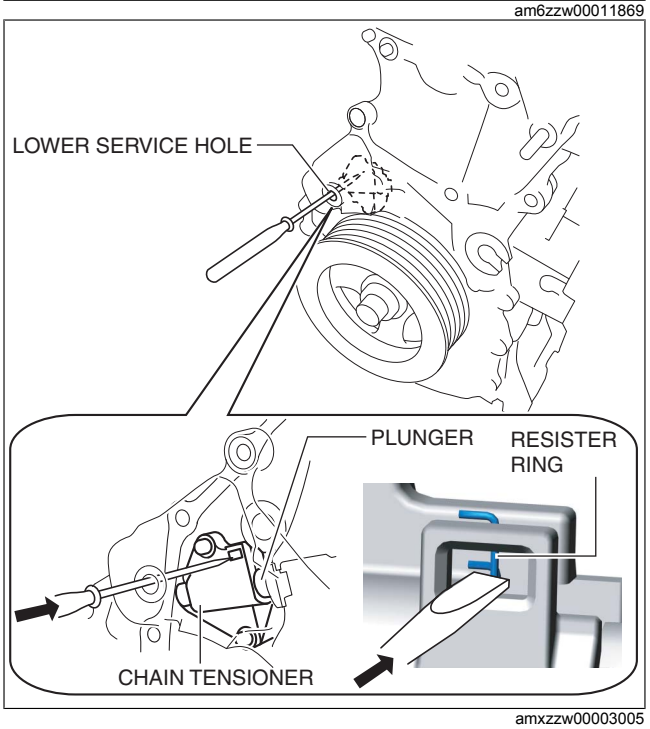


am6zzw00011869

(7) Insert a thin flathead screwdriver (precision screwdriver) into the lower service hole and press the chain tensioner resister ring. (Chain tensioner (type B))

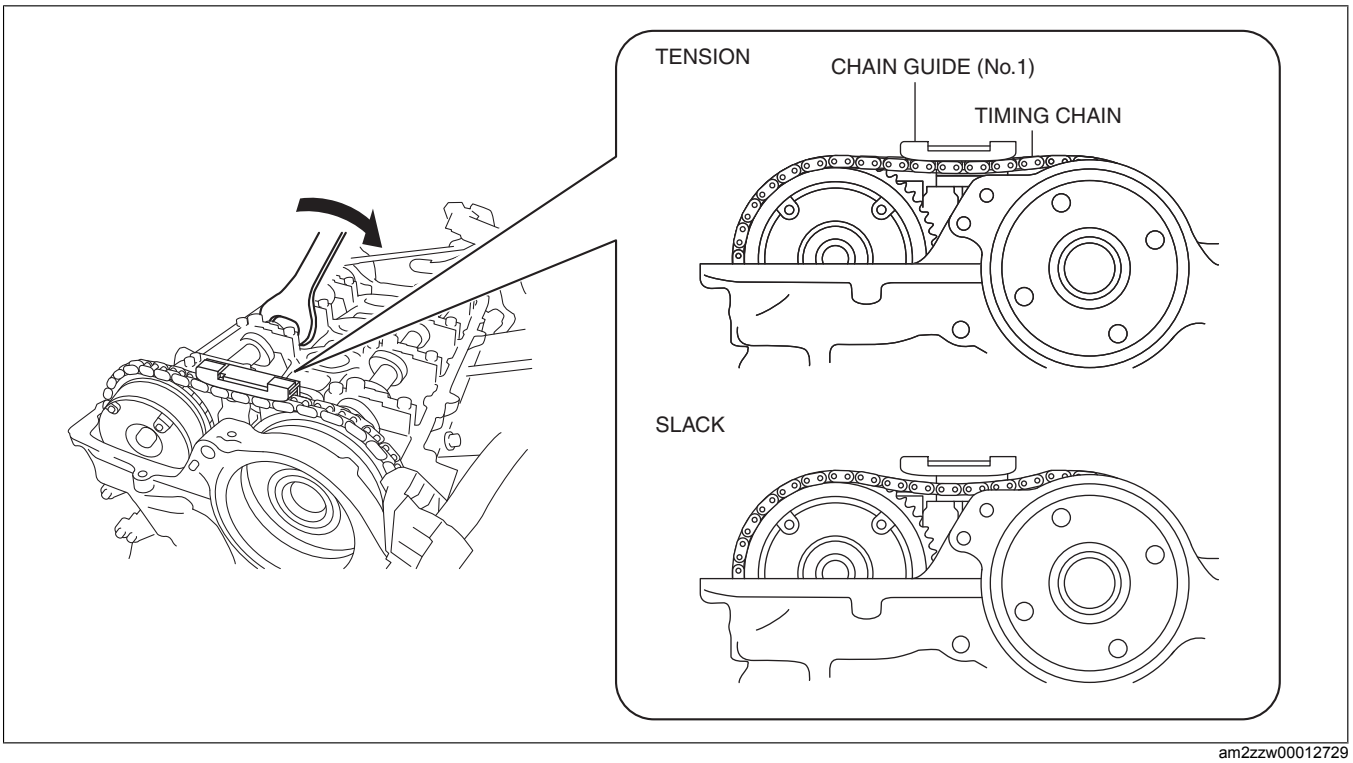
Caution
- **If excessive force is applied to the resister ring, it may cause the resister ring to deform. When pressing the resister ring, be careful not to apply excessive force.**

Note
- The plunger lock is released by pressing the chain tensioner resister ring.



amxzzw00003005

(8) With the plunger lock released, rotate the exhaust camshaft clockwise until the timing chain loosens.

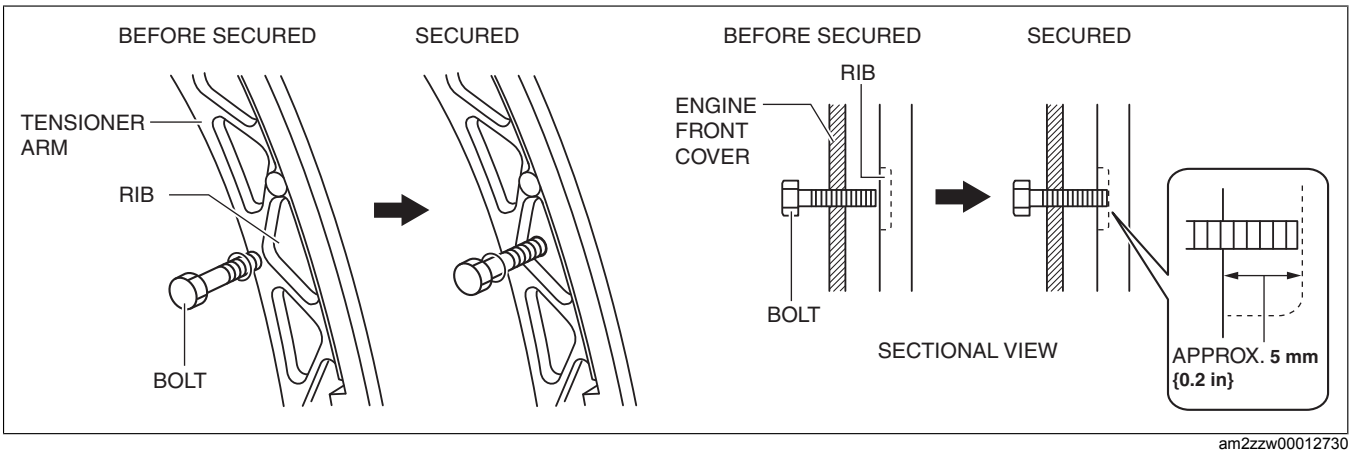


am2zzw00012729

(9) With the chain loosened, tighten the M6 bolt in the upper service hole so that the bolt is inserted **approx. 5 mm {0.2 in}** additionally to secure the tensioner arm.

Note

- By rotating the exhaust camshaft clockwise, the timing chain pushes the tensioner arm and the rib position deviates. The entire tensioner arm can be secured by hooking the bolt onto the deviated rib.



am2zzw00012730

- If the bolt cannot be inserted **approx. 5 mm {0.2 in}**, the plunger lock of the chain tensioner may not release, or the chain may not loosen enough. Return the bolt to the original position and restart the procedure from Step (3).

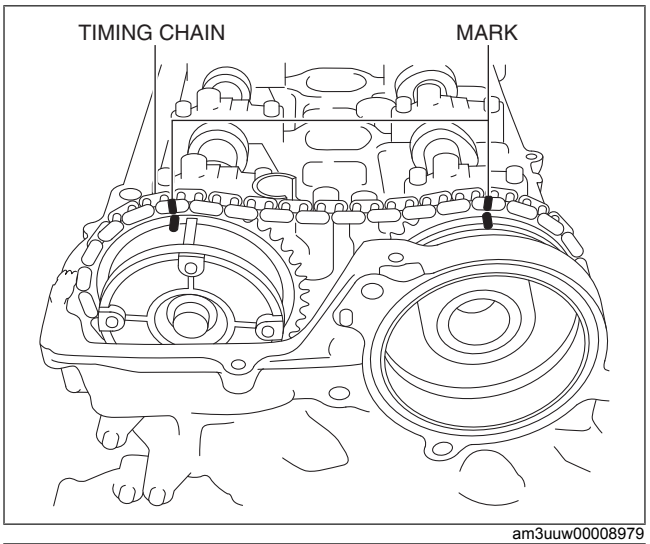
(10) After securing the tensioner arm, remove the chain guide (No.1).

Electric variable valve timing actuator removal note

Caution

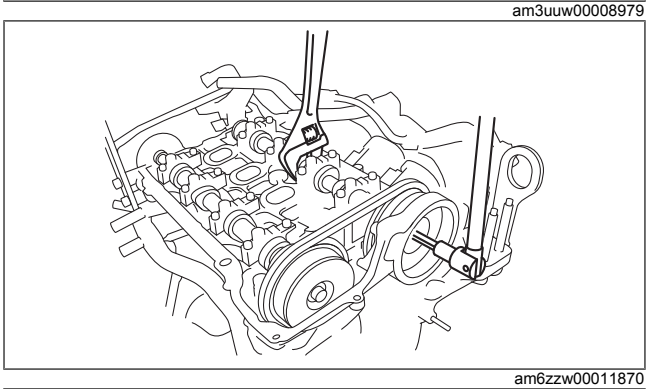
- **Keep the timing chain pulled up during and after the work in which it has been set aside from the sprocket. If the chain falls down, it may disengage from the sprocket on the crankshaft side and deviate, leading to valve timing deviation.**
- **After servicing, always verify the valve timing.**

1. Mark the timing chain and each actuator so that they can be reassembled in the same condition as before removal.

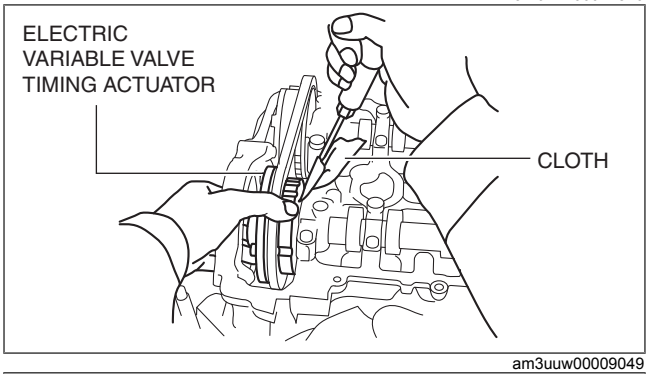


am3uuw00008979

2. Hold the intake camshaft using a wrench on the cast hexagon and remove the electric variable valve timing actuator installation bolt.

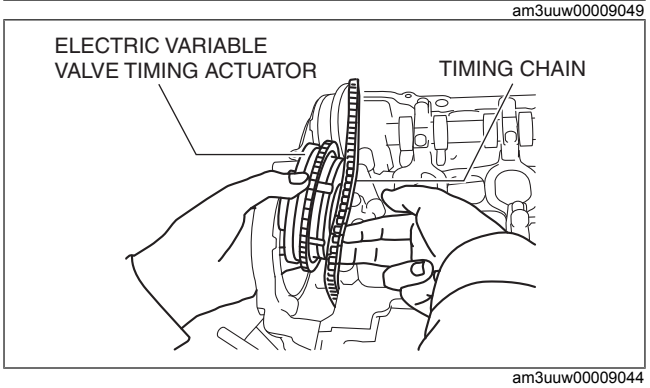

am6zzw00011870

3. Using a screwdriver wrapped in a clean cloth, lightly press the electric variable valve timing actuator to the front and separate the intake camshaft and the actuator.



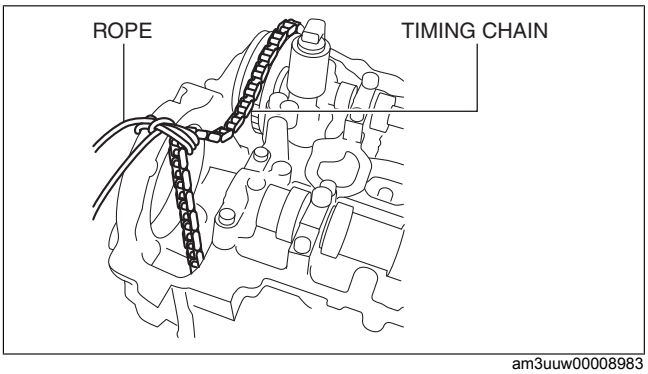
am3uuw00009049

4. Remove the timing chain from the exhaust and intake camshaft sprockets and set is aside, and remove the electric variable valve timing actuator.



am3uuw00009044

5. Return the timing chain to the exhaust camshaft sprocket, and pull the timing chain up and suspend it using a rope as shown in the figure.



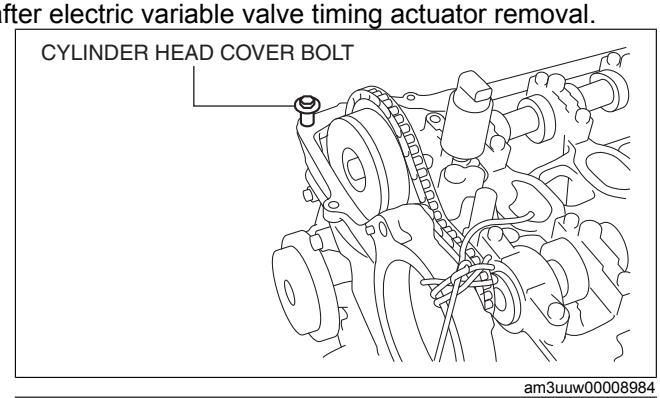
am3uuw00008983

Camshaft removal note

Caution
- **Keep the timing chain pulled up during and after the work in which it has been set aside from the sprocket. If the chain falls down, it may disengage from the sprocket on the crankshaft side and deviate, leading to valve timing deviation.**
- **After servicing, always verify the valve timing.**

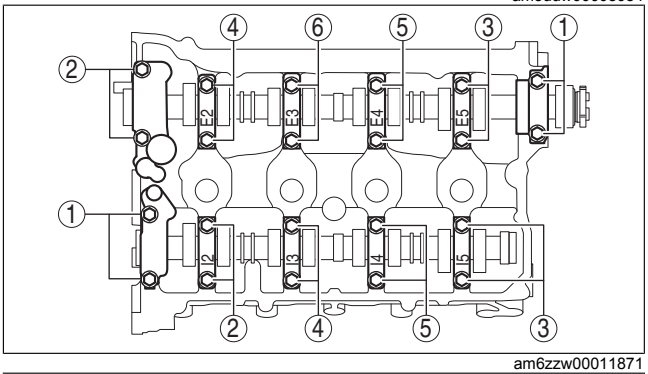
Note
- Remove the exhaust camshaft and the hydraulic variable valve timing actuator as a single unit.

1. Leave the timing chain as it is, which was suspended after electric variable valve timing actuator removal.
2. Temporarily install the cylinder head cover bolt to the position shown in the figure to use it for suspending the timing chain.



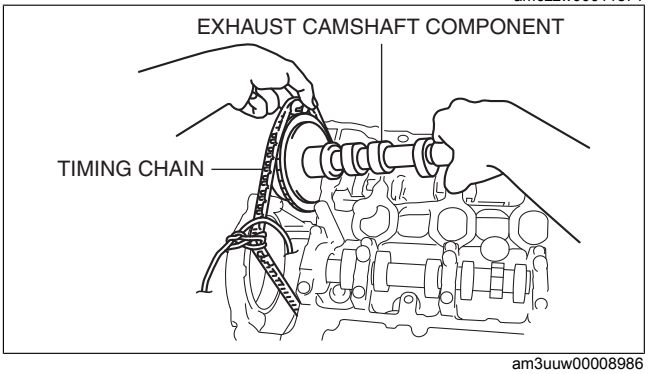
am3uuw00008984

3. Loosen the camshaft cap bolts in two or three passes in the order shown in the figure, and remove the camshaft caps.
4. Remove the intake camshaft.



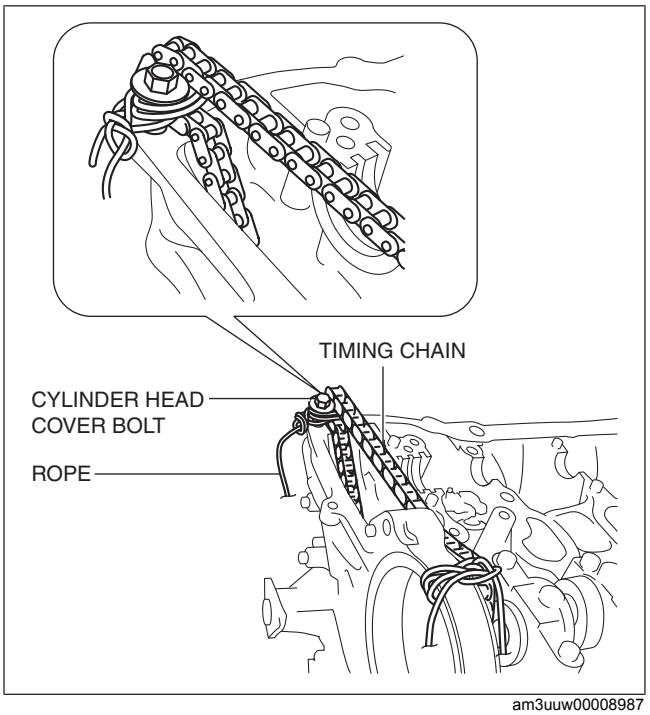
am6zzw00011871

5. While setting the timing chain aside, remove the hydraulic variable valve timing actuator and the exhaust camshaft as a single unit.



am3uuw00008986

6. Pull up the timing chain and suspend it using a rope as shown in the figure.



am3uuw00008987

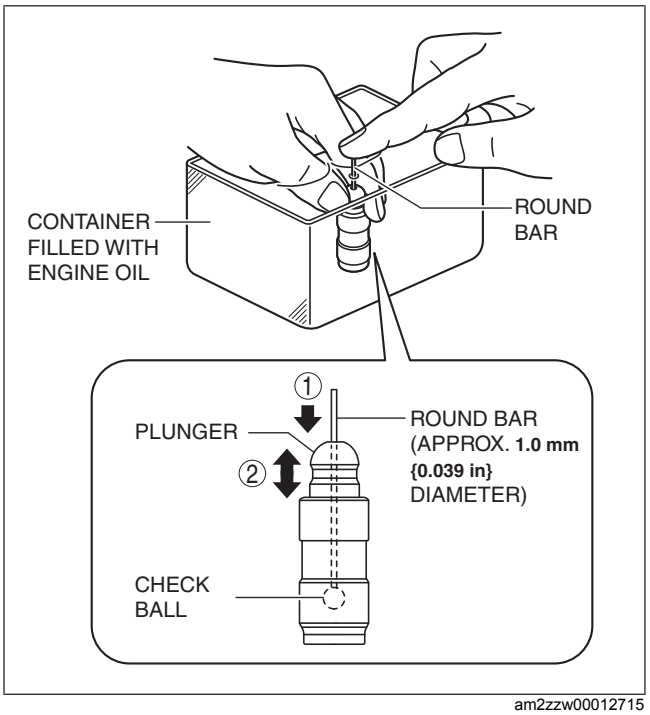
Rocker arm removal note
1. Keep the rocker arms in the order of removal to enable reassembly in their original positions.

HLA removal note
1. Keep the HLAs in the order of removal to enable reassembly in their original positions.

HLA installation note
1. Perform HLA air bleeding using the following procedure.
   (1) Put the HLA in a container filled with engine oil.

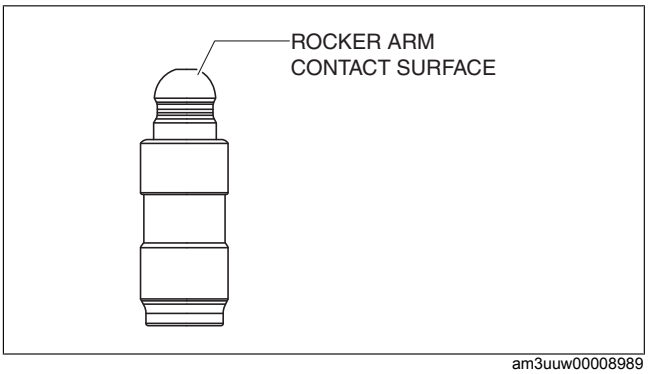
Caution
- **Do not insert the round bar firmly because the check ball spring force is extremely weak.**

   (2) While lightly pressing the check ball using a round bar (**approx. 1.0 mm {0.039 in}** diameter), bleed air by moving the plunger up and down.
   (3) Press the end of the plunger in the oil and verify that there is no rebounding feel.
      - If rebounding feel cannot be eliminated, replace the HLA.



am2zzw00012715

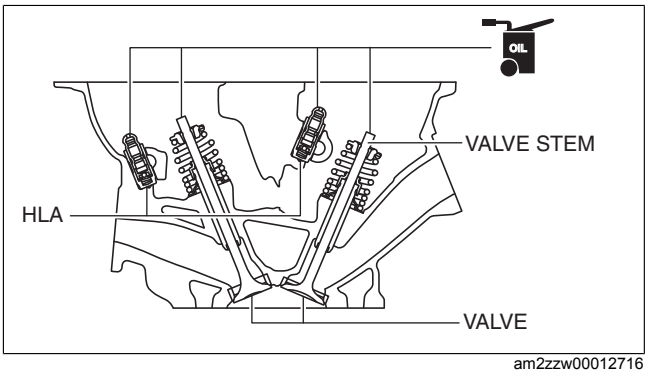
2. Visually inspect the HLA surface where it contacts the rocker arm for wear or damage.
   - If there is any malfunction, replace the HLA.
3. Assemble the HLAs to their original positions.



am3uuw00008989

Rocker arm installation note

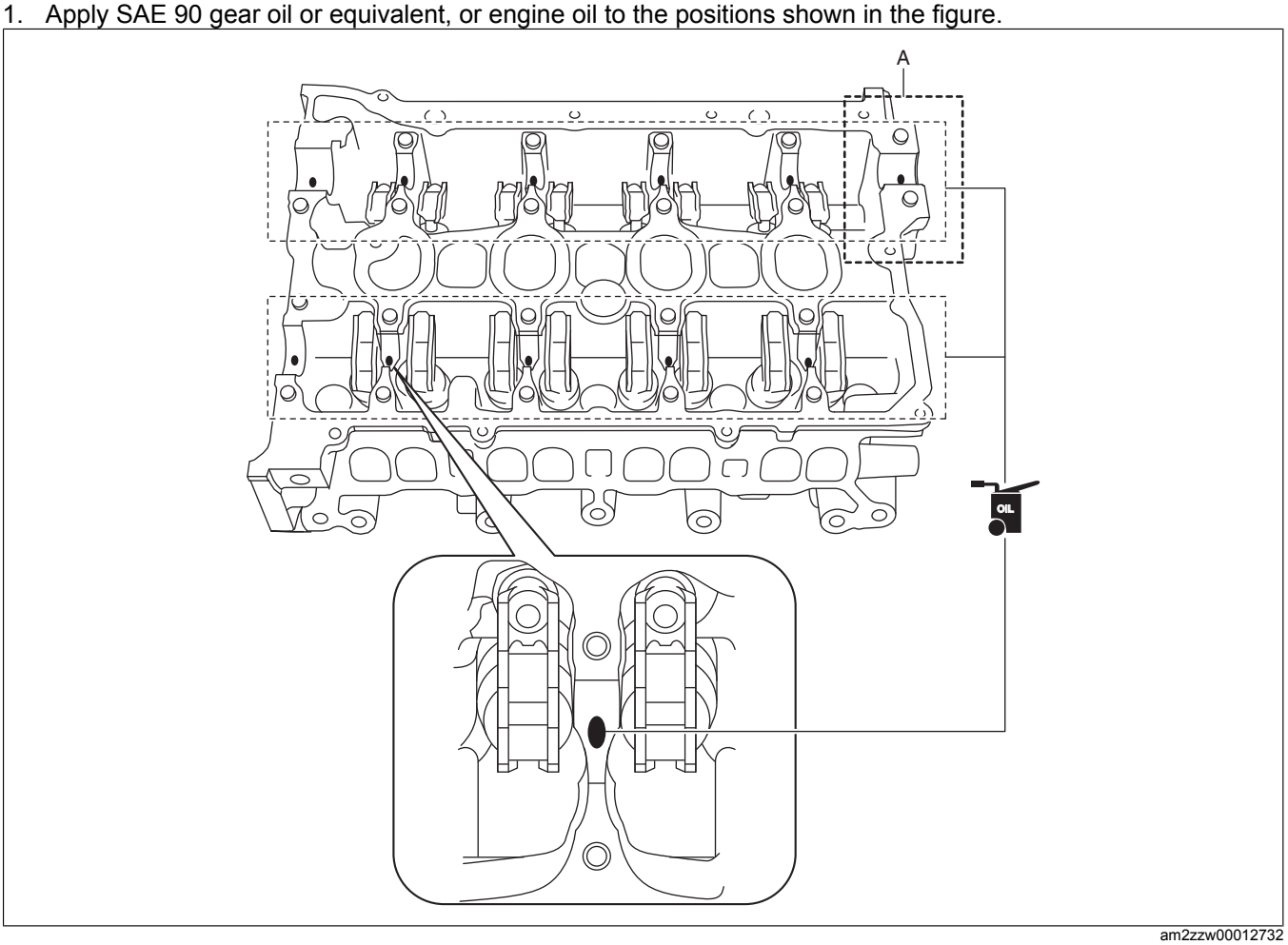
1. Apply engine oil to the HLAs and the end of the valve stems.
2. Install the rocker arms to the same positions as before removal.



am2zzw00012716

Camshaft installation note

1. Apply SAE 90 gear oil or equivalent, or engine oil to the positions shown in the figure.

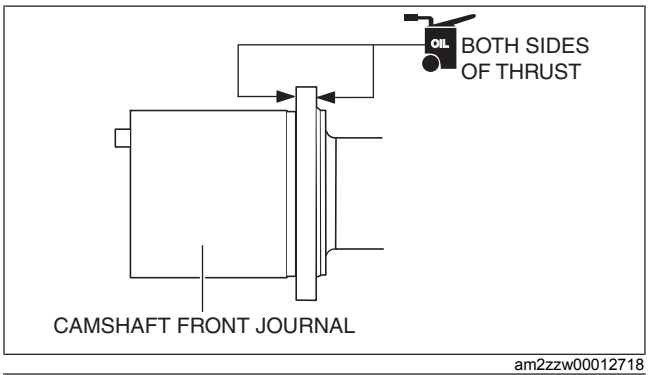


am2zzw00012732

Caution

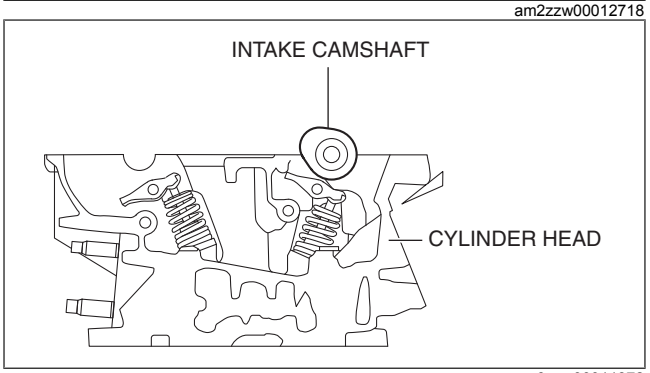
- **Apply 0.05 ml {0.05 cc, 0.003 in^3} or less of oil to area A in the figure.**

2. Apply SAE 90 gear oil or equivalent, or engine oil to the positions shown in the figure.



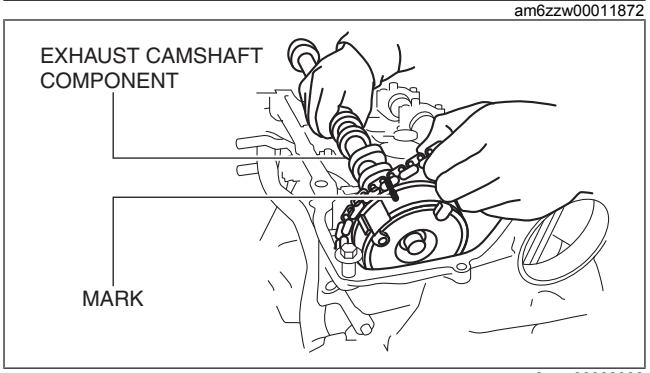
am2zzw00012718

3. As shown in the figure, align the cam position of cylinder No.1 around top dead center (TDC) and place the intake camshaft on the cylinder head.
4. Remove the rope suspending the timing chain on the exhaust side.



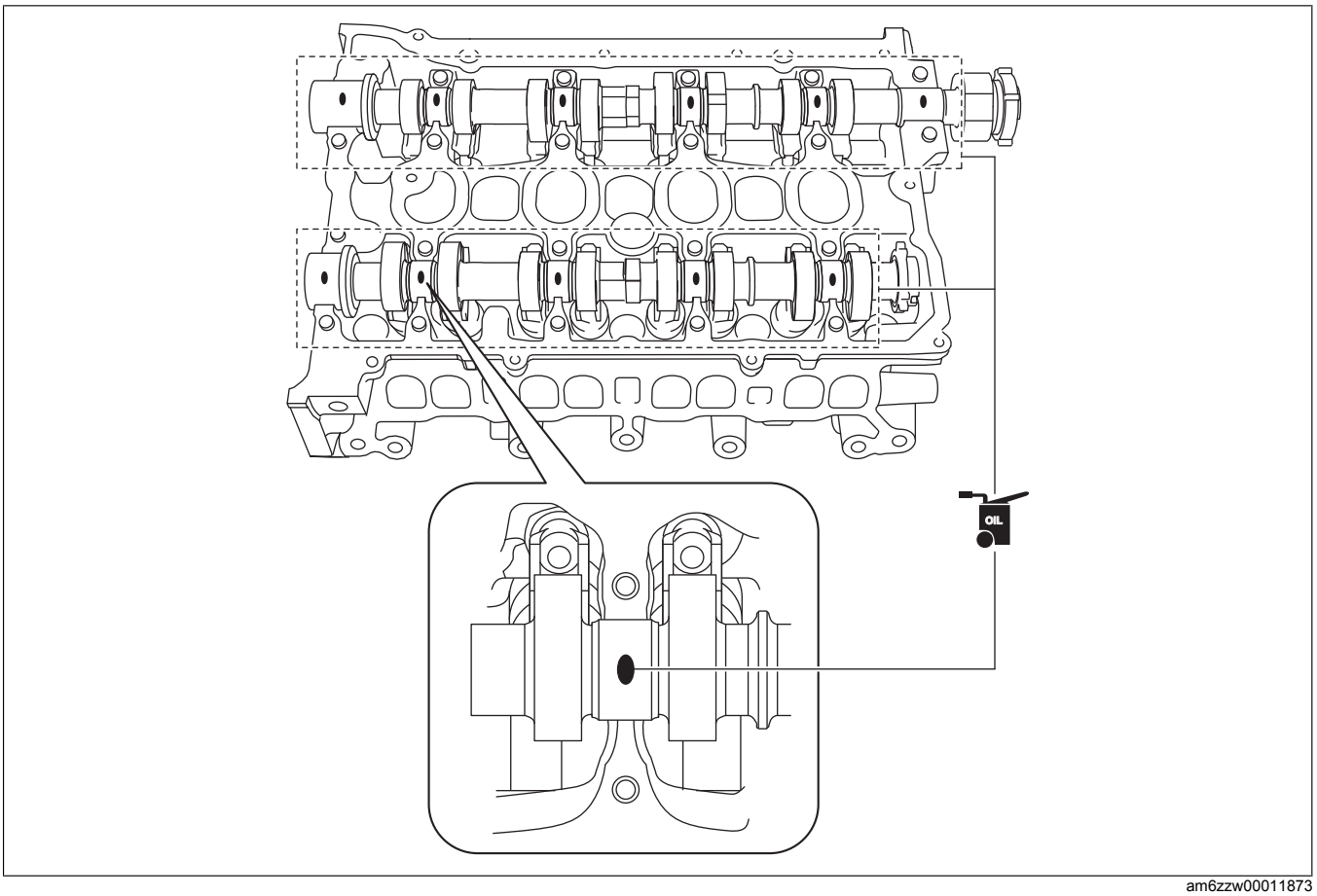
am6zzw00011872

5. While aligning the mark on the timing chain placed during removal with the one on the hydraulic variable valve timing actuator, place the exhaust camshaft component on the cylinder head.



am3uuw00008993

6. Apply SAE 90 gear oil or equivalent, or engine oil to the central area of each journal on the camshaft.



am6zzw00011873

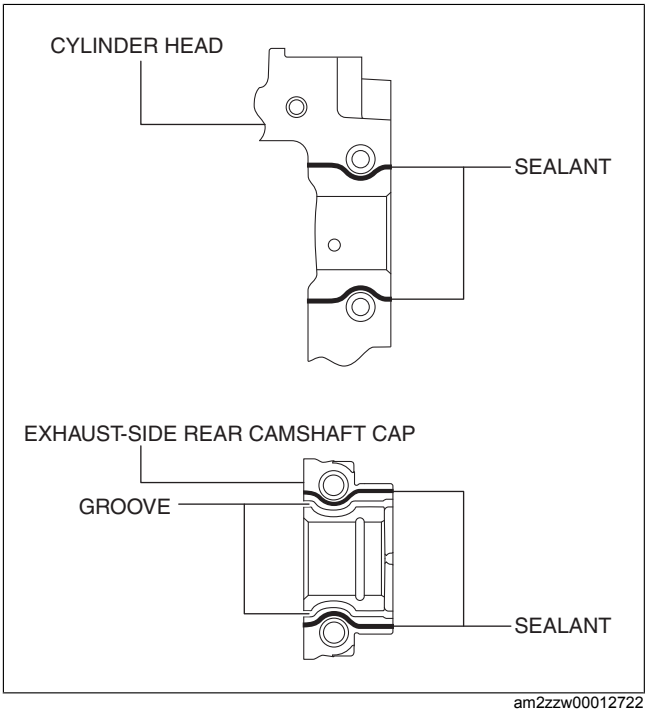
7. Apply sealant (Loctite #962T or equivalent) to the rear camshaft cap installation area on the exhaust side or the rear camshaft caps on the exhaust side of the cylinder head.

Note
- To prevent engine oil leakage, apply sealant to the rear camshaft cap installation area on the exhaust side or the rear camshaft caps on the exhaust side of the cylinder head, and seal the journal.

Caution
- **Do not spill sealant on the journal.**

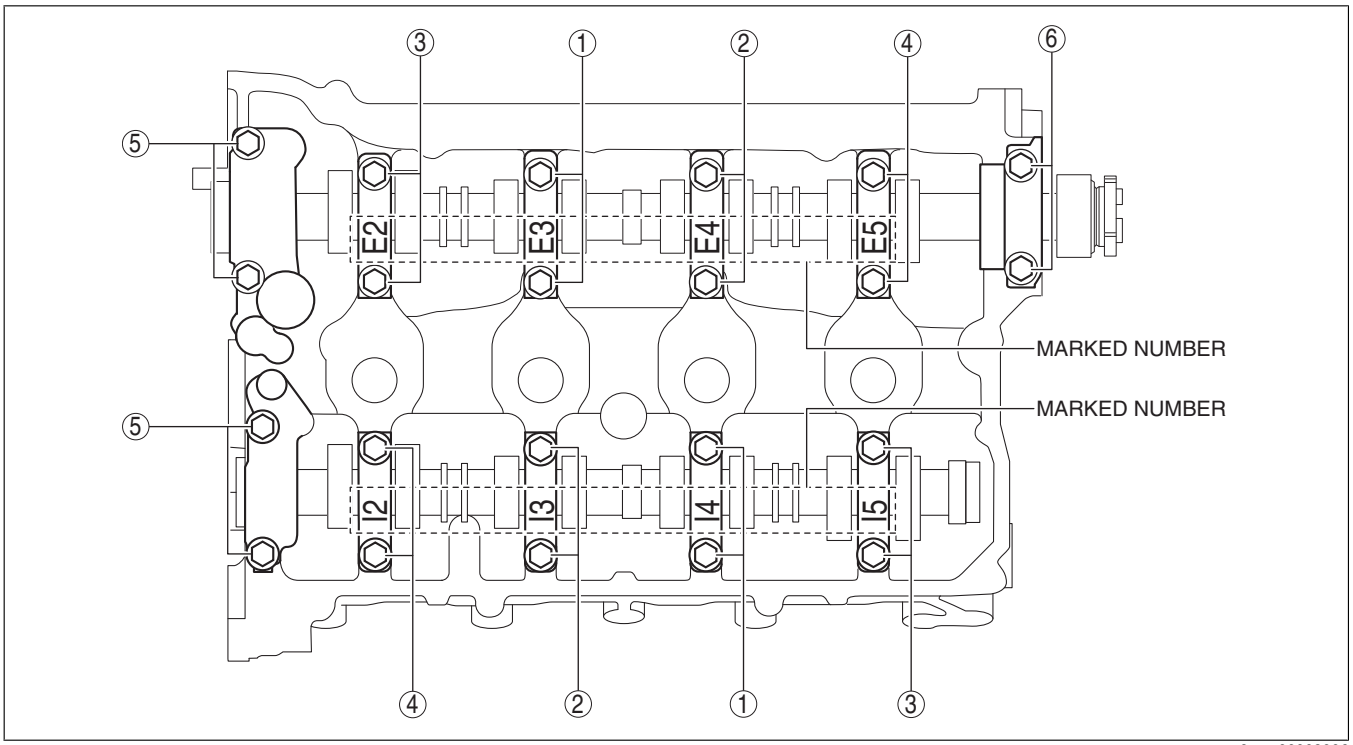
Sealant bead width
0.5—1.5 mm {0.02—0.05 in}

8. Install the camshaft caps in the order of the marked numbers, and temporarily tighten the camshaft cap installation bolts evenly in **2—3 rounds**.



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9. Tighten the camshaft cap installation bolts in two steps in the order shown in the figure.

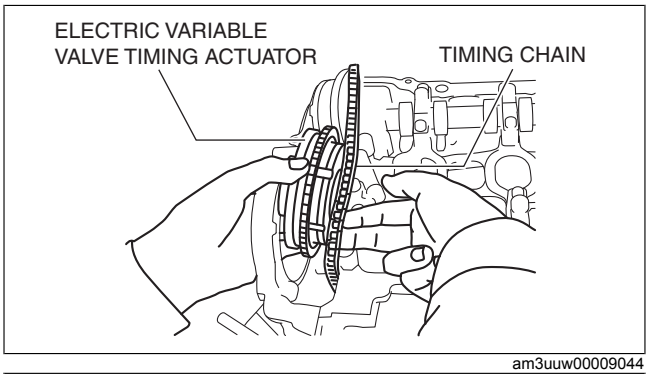


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Tightening torque
Step 1: 3.0—6.0 N·m {31—61 kgf·cm, 27—53 in·lbf}
Step 2: 8—11 N·m {82—112 kgf·cm, 71—97 in·lbf}

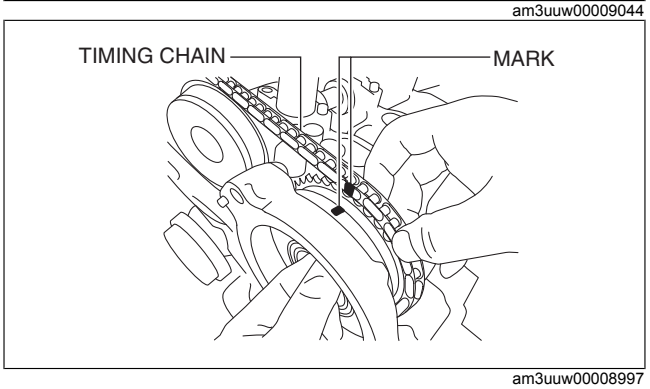
Electric variable valve timing actuator installation note

1. Remove the rope suspending the timing chain on the intake side.
2. Remove the timing chain from the exhaust and intake camshaft sprockets and set it aside to the rear, then insert the electric variable valve timing actuator between the engine front cover and the cylinder head.
3. Return the timing chain to the exhaust camshaft sprocket.



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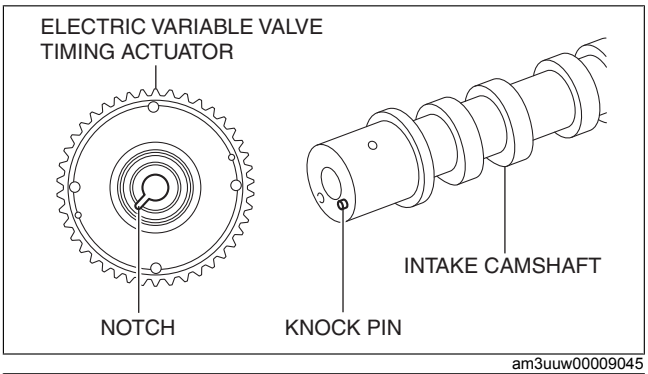
4. While aligning the mark on the timing chain placed during removal with the one on the electric variable valve timing actuator, engage the chain with the sprocket.



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5. Align the knock pin on the end of the intake camshaft with the notch on the actuator side, then assemble the actuator.

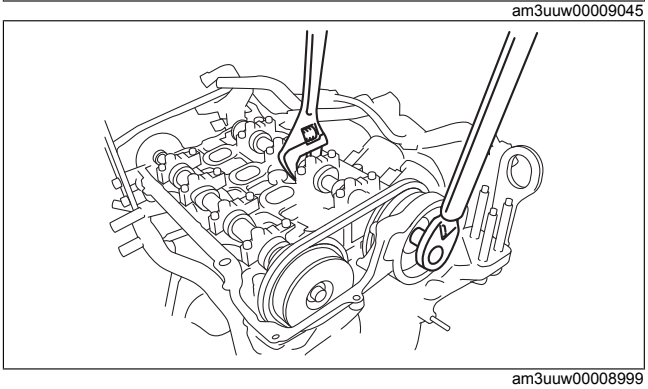
Note
- Adjust the knock pin with the notch while adjusting the knock pin position by rotating the camshaft.



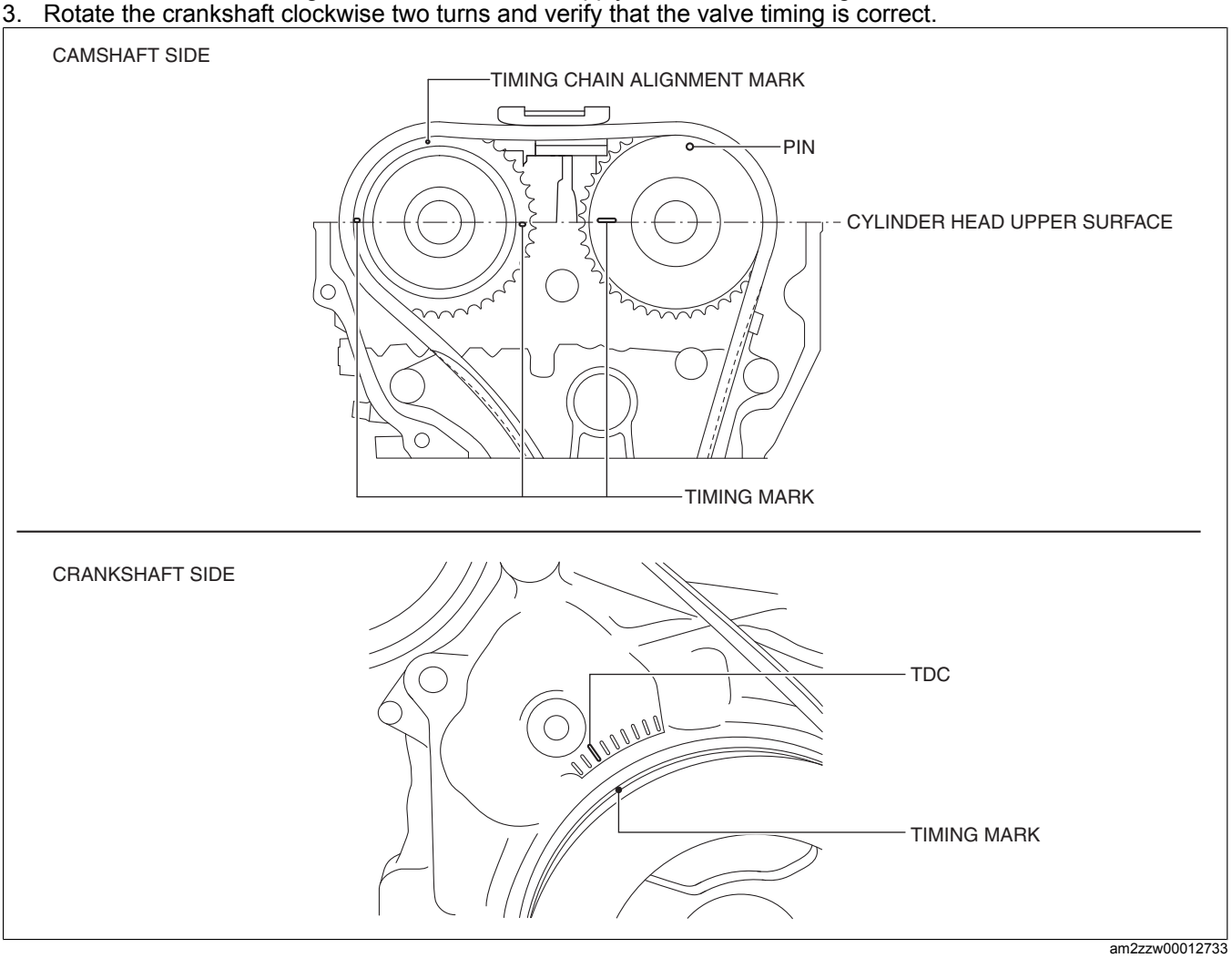
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6. Hold the intake camshaft using a wrench on the cast hexagon, and tighten the electric variable valve timing actuator installation bolt.

Tightening torque
102—114 N·m {10.5—11.6 kgf·m, 76—84 ft·lbf}


am3uuw00008999

Timing chain installation note
1. Install the chain guide (No.1).
2. Remove the bolt securing the tensioner arm and apply tension to the timing chain.
3. Rotate the crankshaft clockwise two turns and verify that the valve timing is correct.



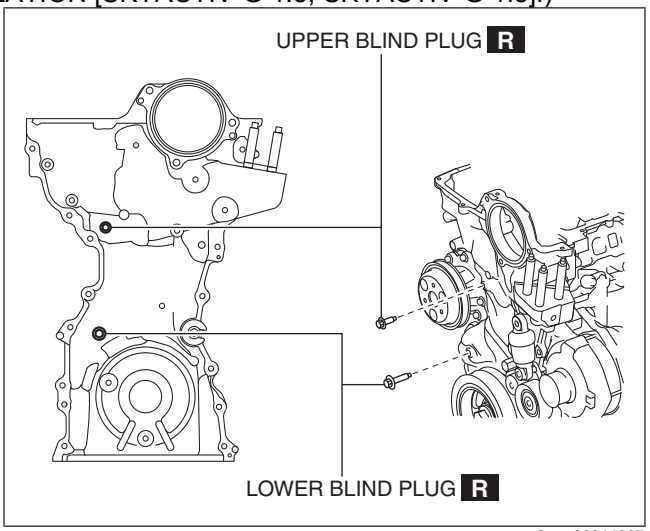
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Note
- The timing mark is not completely parallel with the upper surface of the cylinder head.

- If the valve timing is not correct, remove the engine front cover and reassemble the timing chain to each sprocket. (See TIMING CHAIN REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)

4. Install the blind plugs (upper and lower) for the engine front cover service holes.

Tightening torque
Upper: 8—11 N·m {82—112 kgf·cm, 71—97 in·lbf}
Lower: 17—23 N·m {1.8—2.3 kgf·m, 13—16 ft·lbf}

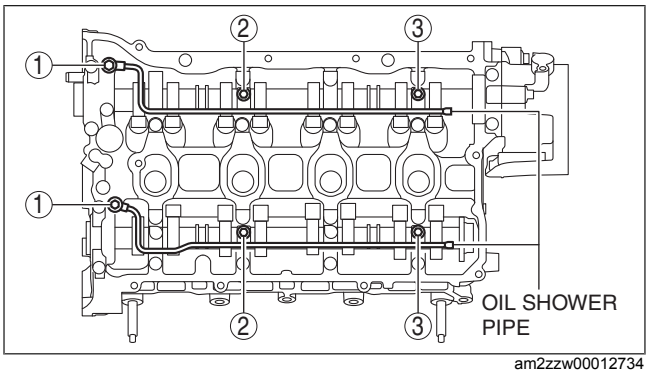


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Oil shower pipe installation note
1. Install the oil shower pipe in the order shown in the figure.

Tightening torque

Installation position	Tightening torque
1	9—12 N·m {92—122 kgf·cm, 80—106 in·lbf}
2, 3	8.0—9.0 N·m {82—91 kgf·cm, 71—79 in·lbf}



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